MIL-M-28787/7C(EC)
7 September 1982
SUPERSEDING
MIL-M-28787/7B(EC)
14 April 1980

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

DECODER, BINARY, DIGITAL, KEY CODE GDN

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic binary decoder module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specificed in MIL-STD-1389

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Naval Electronic Systems Command, ATTN 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.8 failures per million operating hours for class I or 3.27 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.4).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.4).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be GUN.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5 6 Depth. The module shall have no additional depth requirements.

3.5 7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes

3.5.8 Marking Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/7-1 for class I and M28787/7-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table 1.

3.8 End-of-life limits Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2)

3.9 Power supply voltage The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0 5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5 5 Vdc ±1 percent.

4 QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665 Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified

4 4 Test procedures. The module shall be tested as specified in table I

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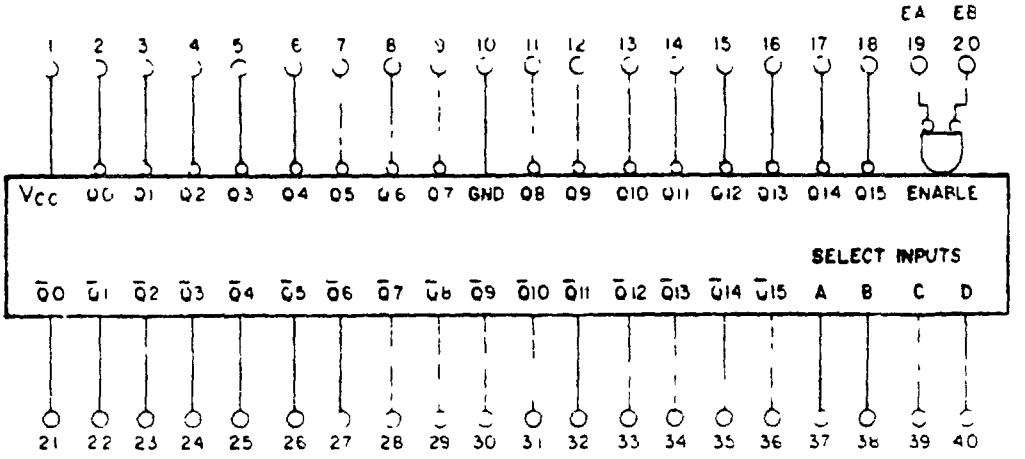


FIGURE 1. GDN module function diagram, binary decoder.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A32	V_{OUT} (low-loaded)	III	S	100	800	100	800	100	800	100	800	100	800	mV
B1-B6	Input breakdown current	IV	S	0	1	0	1	0	1	0	1	0	1	mA
C1-C6	Input clamp diode voltage	V	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
D1-D6	Input current, high level	VI	X	0	40	0	40	0	40	0	60	0	60	µA
E1-E6	Input current, low level	VI	X	0.10	1.73	0.10	1.73	0.10	1.73	0.10	1.73	0.10	1.73	mA
F1-F16	Output short circuit current	VI	X	20	72	20	72	20	72	20	72	20	72	mA
F17-F32	Output short circuit current	VI	X	20	100	20	100	20	100	30	100	30	100	mA
G1-G288	Output voltage, low level	VII	X	0	400	0	400	0	400	0	400	0	400	mV
H1-H288	Output voltage, high level	VII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
J	Power supply current	VIII	X	80	210	80	210	80	210	80	210	80	210	mA
K1-K32	Transition time (t_{TLH}), C_L = 130 pF													
	Initial	IX	S	5	50	5	55	5	55	5	55	5	55	ns
	End-of-life	IX	S	5	55	5	60	5	60	5	60	5	60	ns
L1-L32	Transition time (t_{THL}), C_L = 130 pF													
	Initial	IX	S	5	25	5	30	5	30	5	30	5	30	ns
	End-of-life	IX	S	5	30	5	35	5	35	5	35	5	35	ns

See footnotes at end of table.

* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
M1-M31 (odd numbers)	Propagation delay time (t_{PHL}), C_L = 130 pF													
	Initial	X	S	5	32	5	32	5	32	5	52	5	52	ns
	End-of-life	X	S	5	32	5	32	5	32	5	57	5	57	ns
M2-M32 (even numbers)	Propagation delay time (t_{PHL}), C_L = 130 pF													
	Initial	X	S	5	49	5	49	5	49	5	75	5	75	ns
	End-of-life	X	S	5	49	5	49	5	49	5	80	5	80	ns
M33-M40	Propagation delay time (t_{PHL}), C_L = 130 pF													
	Initial	X	S	5	38	5	38	5	38	5	38	5	38	ns
	End-of-life	X	S	5	38	5	38	5	38	5	43	5	43	ns
N1-N31 (odd numbers)	Propagation delay time (t_{PLH}), C_L = 130 pF													
	Initial	X	S	5	32	5	32	5	32	5	47	5	47	ns
	End-of-life	X	S	5	32	5	32	5	32	5	52	5	52	ns
N2-N32 (even numbers)	Propagation delay time (t_{PLH}), C_L = 130 pF													
	Initial	X	S	5	65	5	65	5	65	5	70	5	70	ns
	End-of-life	X	S	5	65	5	65	5	65	5	75	5	75	ns
N33-N40	Propagation delay time (t_{PLH}), C_L = 130 pF													
	Initial	X	S	5	38	5	38	5	38	5	43	5	43	ns
	End-of-life	X	S	5	38	5	38	5	38	5	48	5	48	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
P	Power supply transient rise time	XI	S	2.5	7.5	2.5	7.5	2.5	7.5	2.5	7.5	2.5	7.5	µs
Q	Isolation, frame to module pins													
	Initial	XII	S	10		10		10		10		10		MΩ
	End-of-life	XII	S	1		1		1		1		1		MΩ
R1-R112	Output crosstalk, low level	XIII	Q	0	400	0	400	0	400	0	400	0	400	mV
S1 S112	Output crosstalk, high level	XIII	Q	0	400	0	400	0	400	0	400	0	400	mV
T	Power supply transient amplitude	XIV	Q	0	400	0	400	0	400	0	400	0	400	mV
U1-U6	Input capacitance	XV	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through U shall be performed in any order with the following exceptions. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows A, B where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through U in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through Q shall be performed in the order specified in 4 4.2 using the initial limits for applicable operating temperature At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4 Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2 Following the life test and while still at the maximum class temperature, electrical tests C through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I Following the return of the module to 25°C, electrical tests C through U shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I

5 PACKAGING

5 1 Packaging requirements The requirements for packaging shall be in accordance with MIL-M-28787

6 NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6 2 End-of-life When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification Military Specification MIL-M-28787/7 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No 10001-2656675 Part number M28787/7-1 supersedes part numbers 2658675 and M28787/7

6.4 Substitutability Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3 3 and 3.4)

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* 6.5 Module function. This module contains a 1 of 16 decoder which accepts 4 active high select inputs (A, B, C, and D) and provides 16 mutually exclusive active low-true (Q) outputs and 16 mutually exclusive active high-complementary ($\overline{Q}$) outputs. The decoder has an active low ANDed enable (EA and EB) The D input produces a useful inhibit function when the GDN module is used as a one out of eight decoder.

6.5.1 Input to output relationship. Each binary decoder module has the input to output relationship of table II for positive logic.

6.5.2 Parallel outputs. Paralleling of outputs is not permissible

6.6 Changes from previous issue The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous isue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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TABLE II Input to output relationship 1/ - 3/

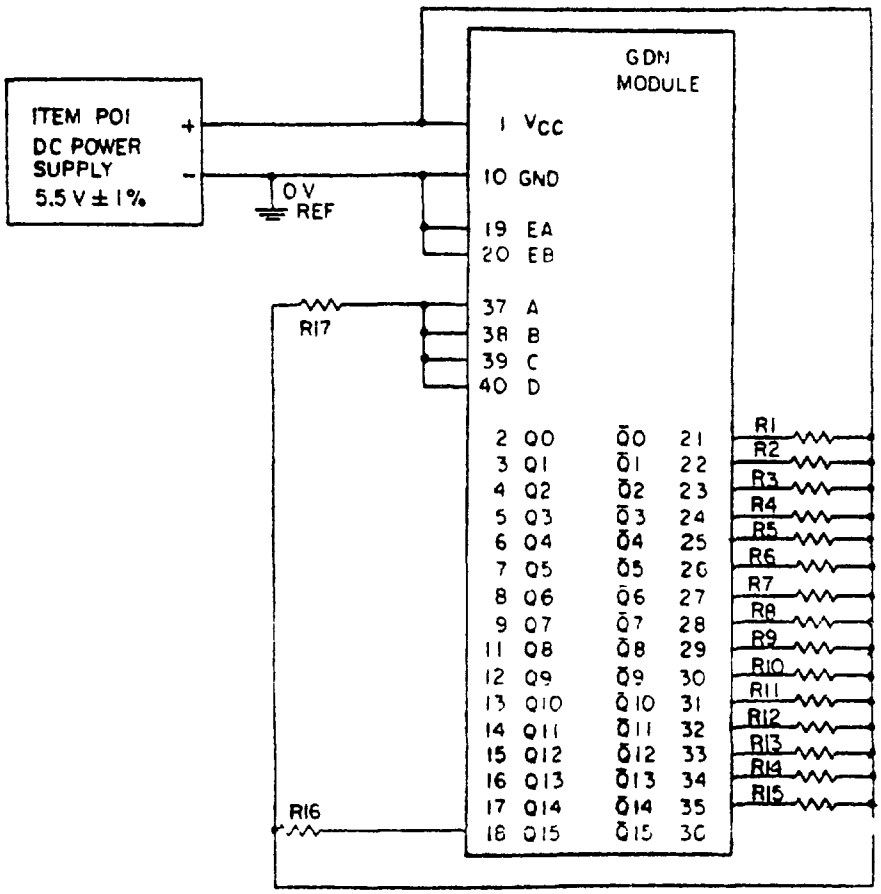
Enable		Select				Q outputs															$\overline{Q}$ outputs															
$\overline{EA}$	$\overline{EB}$	A	B	C	D	Q0	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	$\overline{Q0}$	$\overline{Q1}$	$\overline{Q2}$	$\overline{Q3}$	$\overline{Q4}$	$\overline{Q5}$	$\overline{Q6}$	$\overline{Q7}$	$\overline{Q8}$	$\overline{Q9}$	$\overline{Q10}$	$\overline{Q11}$	$\overline{Q12}$	$\overline{Q13}$	$\overline{Q14}$	$\overline{Q15}$
19	20	37	38	39	40	2	3	4	5	6	7	8	9	11	17	13	14	15	16	17	18	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
H	H		X	X	X	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L
H	L	X	X	X	X	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L
L	H	X	X	X	X	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L
L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L
L	L	H	L	L	L	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	L	L	L	L	L	L	L	L	L	L	L	L	L	L
L	L	L	H	L	L	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	L	L	H	L	L	L	L	L	L	L	L	L	L	L	L	L
L	L	H	H	L	L	H	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H	L	L	L	H	L	L	L	L	L	L	L	L	L	L	L	L
L	L	L	L	H	L	H	H	H	H	L	H	H	H	H	H	H	H	H	H	H	H	L	L	L	L	H	L	L	L	L	L	L	L	L	L	L	L
L	L	H	L	H	L	H	H	H	H	H	L	H	H	H	H	H	H	H	H	H	H	L	L	L	L	L	H	L	L	L	L	L	L	L	L	L	L
L	L	L	H	H	L	H	H	H	H	H	H	L	H	H	H	H	H	H	H	H	H	L	L	L	L	L	L	H	L	L	L	L	L	L	L	L	L
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L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	L	L	L
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L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	L	L	L	L	L	L	L	L	L	L	L	L	L	L	H	L
L	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	H

1/ H = high level

2/ L = low level

3/ X = don't care (high level or low level)

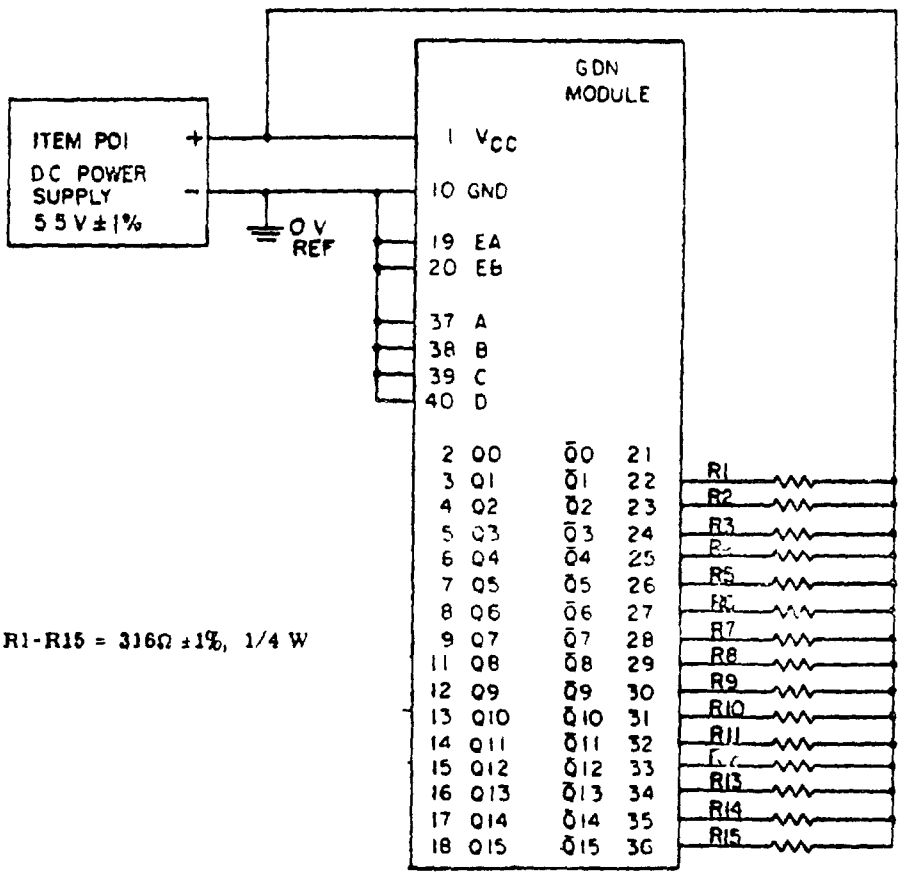
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NOTES
1 R1-R15 = 316Ω ±1%, 1/4 W
2 R16 = R17 = 357Ω ±1%, 1/4 W.

FIGURE 2 Operating temperature and life test circuit

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NOTE R1-R15 = 316Ω ±1%, 1/4 W

FIGURE 3 Component temperature test circuit

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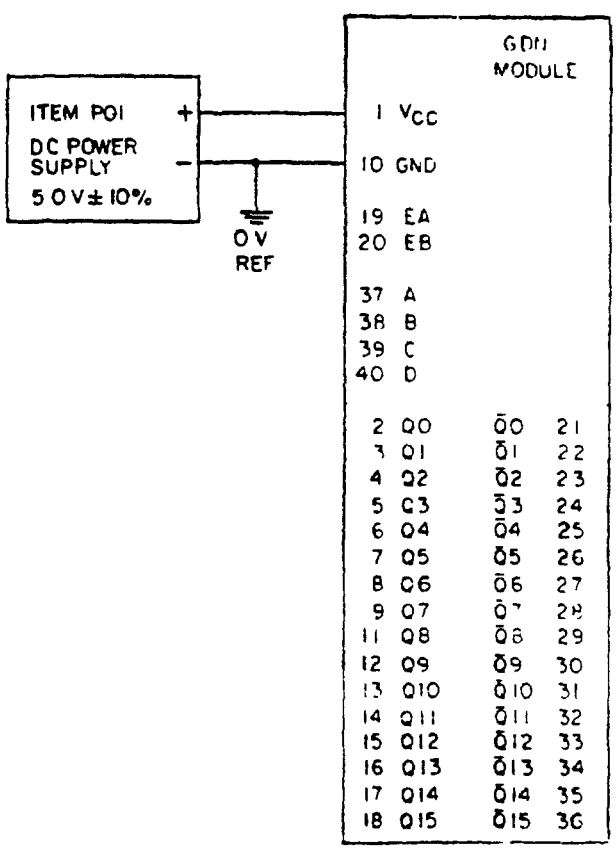


FIGURE 4 Vibration test circuit

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* TABLE III. V_{OUT} (low-loaded) test.

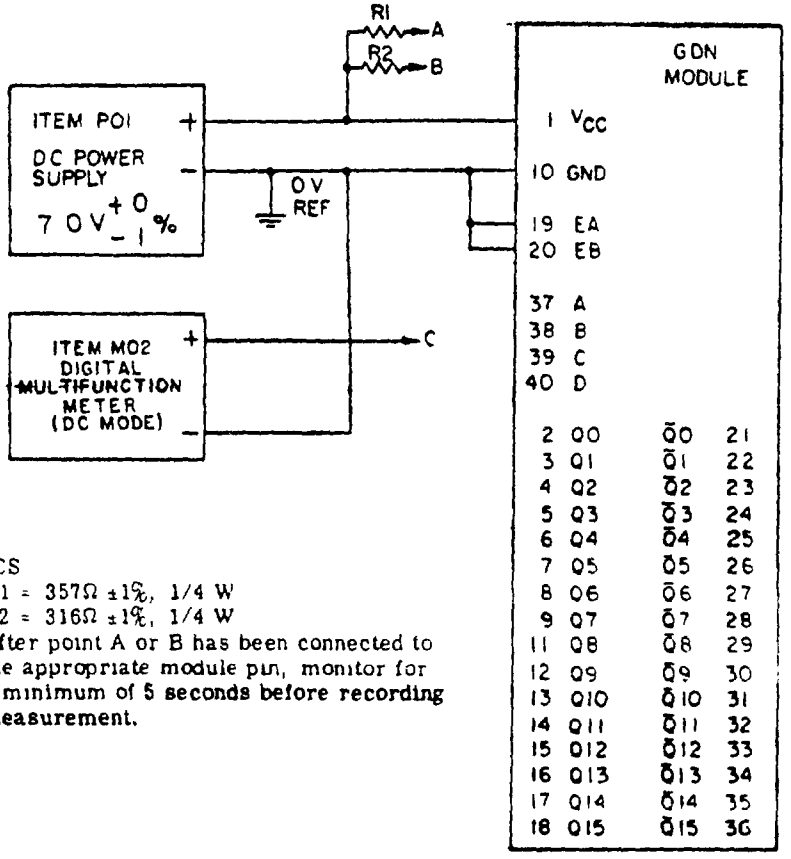
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin 10 to pin	Min	Max		
A1	Q0 V_{OUT} (low loaded)	2	Open	2	37,38,39,40	100	800	mV	5
A2	Q1	3		3	38,39,40				
A3	Q2	4		4	37,39,40				
A4	Q3	5		5	39,40				
A5	Q4	6		6	37,38,40				
A6	Q5	7		7	38,40				
A7	Q6	8		8	37,40				
A8	Q7	9		9	40				
A9	Q8	11		11	37,38,39				
A10	Q9	12		12	38,39				
A11	Q10	13		13	37,39				
A12	Q11	14		14	39				
A13	Q12	15		15	37,38				
A14	Q13	16		16	38				
A15	Q14	17		17	37				
A16	Q15	18	Open	18	Open				
A17	$\overline{Q0}$	Open	21	21	38,39,40				
A18	$\overline{Q1}$	Open	22	22	37,38,39,40				
A19	$\overline{Q2}$	Open	23	23	37,38,39,40				

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* TABLE III. V_{OUT} (low-loaded) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin 10 to pin	Min	Max		
A20	$\overline{Q}3$ V_{OUT} (low loaded)	Open	24	24	37,38,39,40	100	800	mV	5
A21	$\overline{Q}4$		25	25					
A22	$\overline{Q}5$		26	26					
A23	$\overline{Q}6$		27	27					
A24	$\overline{Q}7$		28	28					
A25	$\overline{Q}8$		29	29					
A26	$\overline{Q}9$		30	30					
A27	$\overline{Q}10$		31	31					
A28	$\overline{Q}11$		32	32					
A29	$\overline{Q}12$		33	33					
A30	$\overline{Q}13$		34	34					
A31	$\overline{Q}14$		35	35					
A32	$\overline{Q}15$		36	36					

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NOTES
1 R1 = 357Ω ±1%, 1/4 W
2 R2 = 316Ω ±1%, 1/4 W
3 After point A or B has been connected to the appropriate module pin, monitor for a minimum of 5 seconds before recording measurement.

* FIGURE 5 V_{OUT} (low-loaded) test circuit for table III

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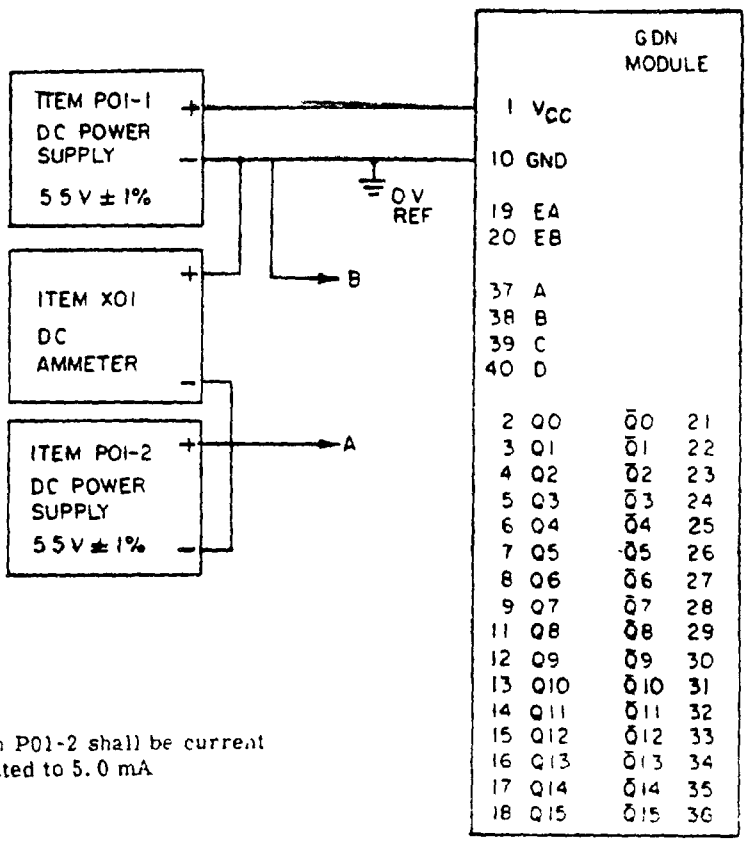
TABLE IV. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Enable A Input breakdown current	19	20	0	1	mA	6
B2	Enable B	20	19				
B3	A Input	37	Open				
B4	B Input	38					
B5	C Input	39					
B6	D Input	40					

TABLE V. Input clamp diode voltage test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
C1	Enable A Input clamp diode voltage	19	-0.4	-1.5	V	7
C2	Enable B	20				
C3	A Input	37				
C4	B Input	38				
C5	C Input	39				
C6	D Input	40				

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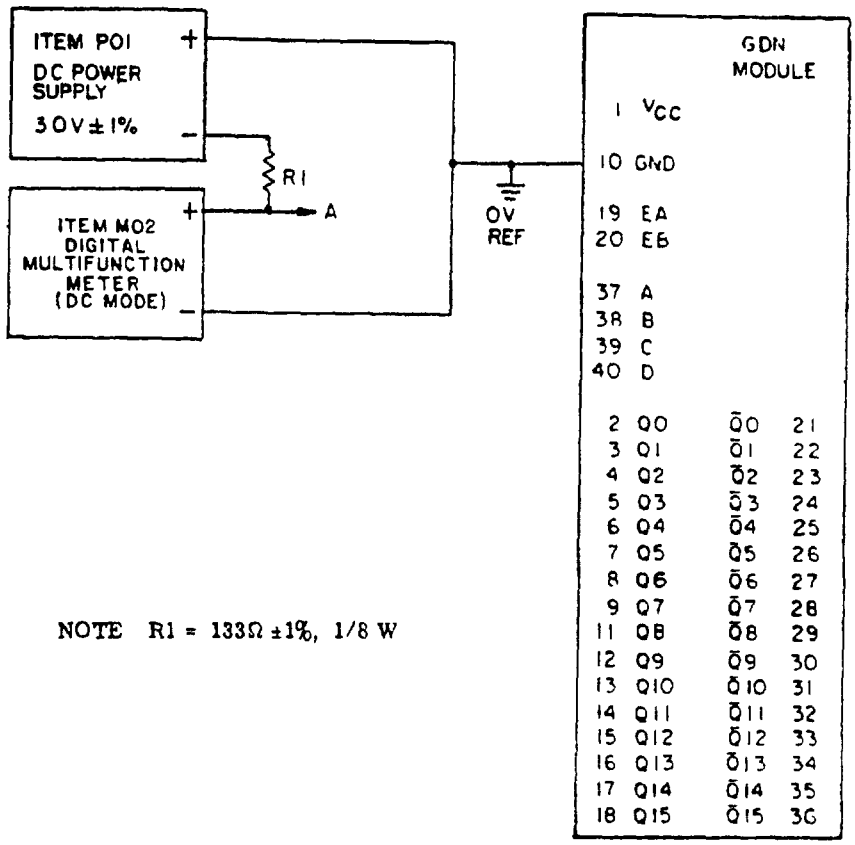
NOTE Item P01-2 shall be current limited to 5.0 mA.

CAUTION

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 6 Input breakdown current test circuit for table IV

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NOTE R1 = 133Ω ±1%, 1/8 W

* FIGURE 7 Input clamp diode voltage test circuit for table V.

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* TABLE VI. Input current (low-level and high-level) and output short circuit current test.

Test condition	Parameter		Connect			Read meter	Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin		Min	Max		
D1	Enable A	Input current, high level	19	Open	20	X01-1	0	40	uA	8
D2	Enable B		20		19					
D3	A Input		37		Open					
D4	B Input		38							
D5	C Input		39							
D6	D Input		40							
E1	Enable A	Input current, low level	20	19		X01-2	0.10	1.73	mA	
E2	Enable B		19	20						
E3	A Input		Open	37						
E4	B Input			38						
E5	C Input			39						
E6	D Input			40						
F1	Q0	Output short circuit current	37	2	19,20,38,39, 40		20	72		
F2	Q1		Open	3	19,20,37,38, 39,40					
F3	Q2			4						
F4	Q3			5						
F5	Q4			6						

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* TABLE VI. Input current (low-level and high-level) and output short circuit current test - Continued.

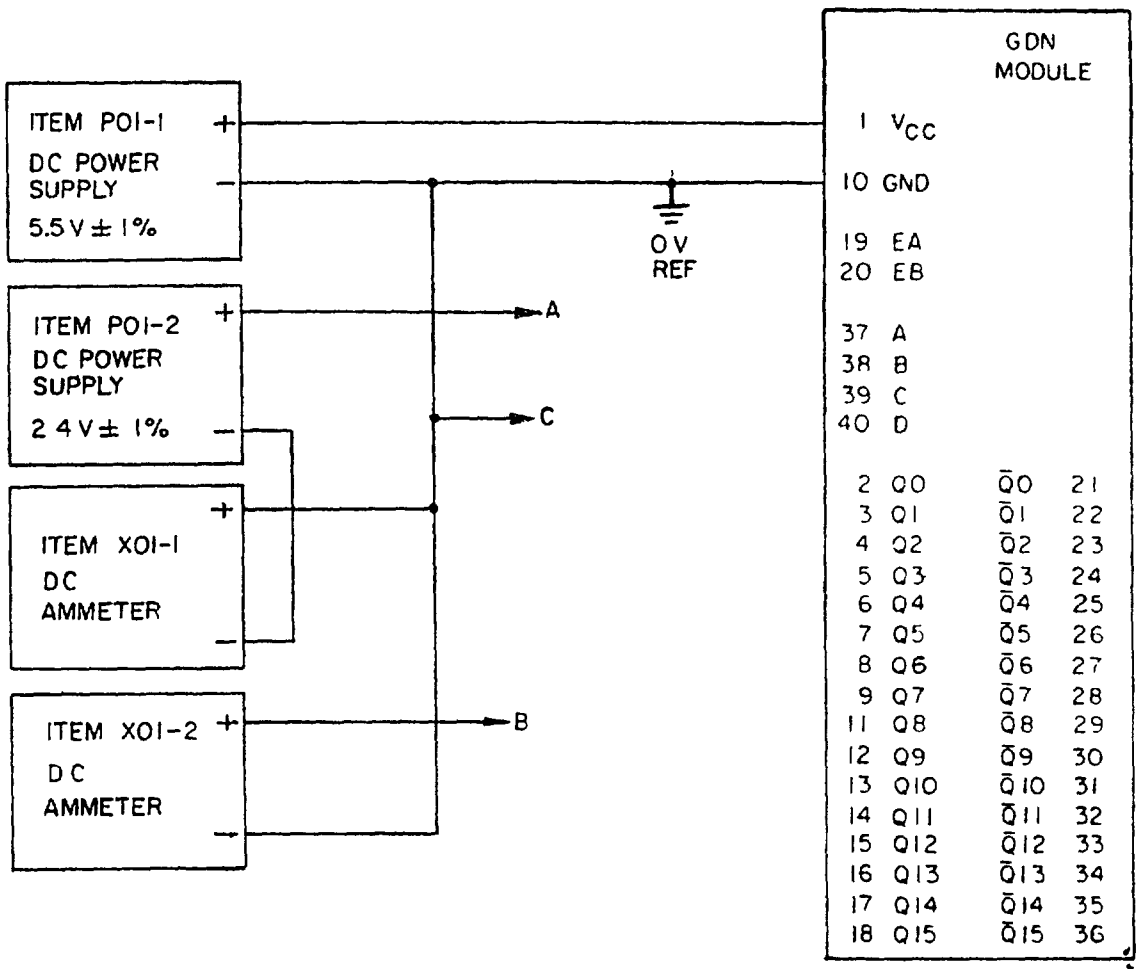
Test condition	Parameter		Connect			Read meter	Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin		Min	Max		
E6	Q5	Output short circuit current	Open	7	19,20,37,38, 39,40	X01-2	20	72	mA	8
F7	Q6			8						
F8	Q7			9						
F9	Q8			11						
F10	Q9			12						
F11	Q10			13						
F12	Q11			14						
F13	Q12			15						
F14	Q13			16						
F15	Q14			17						
F16	Q15			18						
F17	$\overline{Q0}$			21				100		
F18	$\overline{Q1}$		37	22	19,20,38,39, 40					
F19	$\overline{Q2}$		38	23	19,20,37,39, 40					
F20	$\overline{Q3}$		37,38	24	19,20,39,40					
F21	$\overline{Q4}$		39	25	19,20,37,48, 40					
F22	$\overline{Q5}$		37,39	26	19,20,38,40					

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* TABLE VI. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter	Connect			Read meter	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin		Min	Max		
F23	$\overline{Q}6$ Output short circuit current	38,39	27	19,20,37,40	X01-2	20	100	mA	8
F24	$\overline{Q}7$	37,38,39	28	19,20,40					
F25	$\overline{Q}8$	40	29	19,20,37,38, 39					
F26	$\overline{Q}9$	37,40	30	19,20,38,39					
F27	$\overline{Q}10$	38,40	31	19,20,37,39					
F28	$\overline{Q}11$	37,38,40	32	19,20,39					
F29	$\overline{Q}12$	39,40	33	19,20,37,38					
F30	$\overline{Q}13$	37,39,40	34	19,20,38					
F31	$\overline{Q}14$	38,39,40	35	19,20,37					
F32	$\overline{Q}15$	37,38,39,40	36	19,20					

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CAUTION.

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTES:
1. Since items XO1-1 and XO1-2 are not used simultaneously, only one item XO1 is required.
2. With point A disconnected, item XO1-1 should indicate 0.0 current on the 1.0 µA range.
3. When performing the output short circuit current test, only one output shall be shorted at a time, and the measurement taken within 5 seconds.

* FIGURE 8. Input current (low-level and high-level) and output short circuit current test circuit for table VI.

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* TABLE VII. Output voltage (low-level and high-level) (threshold inputs) test.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
G1	Q0 Output voltage (enable low level)	Open	19,20	2	2	21	37,36,39,40	0	400	mV	9
G2	Q1			3	3	22	38,39,40				
G3	Q2			4	4	23	37,39,40				
G4	Q3			5	5	24	39,40				
G5	Q4			6	6	25	37,38,40				
G6	Q5			7	7	26	38,40				
G7	Q6			8	8	27	37,40				
G8	Q7			9	9	28	40				
G9	Q8			11	11	29	37,38,39				
G10	Q9			12	12	30	38,39				
G11	Q10			13	13	31	37,39				
G12	Q11			14	14	32	39				
G13	Q12			15	15	33	37,38				
G14	Q13			16	16	34	38				
G15	Q14			17	17	35	37				
G16	Q15			18	18	36	Open				
G17	$\overline{Q0}$ Output voltage (enable high level)	19	20	21	2	21	38,39,40				
G18	$\overline{Q1}$			22	3	22	37,38,39,40				
G19	$\overline{Q2}$			23	4	23					
G20	$\overline{Q3}$			24	5	24					
G21	$\overline{Q4}$			25	6	25					
G22	$\overline{Q5}$			26	7	26					
G23	$\overline{Q6}$			27	8	27					

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TABLE VII. Output voltage (low-level and high-level) threshold inputs. Dual ...

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
G24	$\overline{Q}7$ Output voltage (enable high level)	19	20	28	9	28	37,38,39,40	0	400	mV	9
G25	$\overline{Q}6$			29	11	29					
G26	$\overline{Q}9$			30	12	30					
G27	$\overline{Q}10$			31	13	31					
G28	$\overline{Q}11$			32	14	32					
G29	$\overline{Q}12$			33	15	33					
G30	$\overline{Q}13$			34	16	34					
G31	$\overline{Q}14$			35	17	35					
G32	$\overline{Q}15$			36	18	36					
G33	$\overline{Q}0$ Output voltage (A, B, C, and D low level)	Open	37,38, 39,40	2	2	21	19,20				
G34	$\overline{Q}1$			22	3	22					
G35	$\overline{Q}2$			23	4	23					
G36	$\overline{Q}3$			24	5	24					
G37	$\overline{Q}4$			25	6	25					
G38	$\overline{Q}5$			26	7	26					
G39	$\overline{Q}6$			27	8	27					
G40	$\overline{Q}7$			28	9	28					
G41	$\overline{Q}8$			29	11	29					
G42	$\overline{Q}9$			30	12	30					
G43	$\overline{Q}10$			31	13	31					
G44	$\overline{Q}11$			32	14	32					
G45	$\overline{Q}12$			33	15	33					
G46	$\overline{Q}13$			34	16	34					
G47	$\overline{Q}14$			35	17	35					
G48	$\overline{Q}15$			36	18	36					

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* TABLE VII. Output voltage (low-level and high-level) (threshold inputs) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
G49	Q1 Output voltage (A high level, B, C, and D low level)	37	38,39, 40	3	3	22	19,20	0	400	mV	9
G50	$\overline{Q}0$			21	2	21					
G51	$\overline{Q}2$			23	4	23					
G52	$\overline{Q}3$			24	5	24					
G53	$\overline{Q}4$			25	6	25					
G54	$\overline{Q}5$			26	7	26					
G55	$\overline{Q}6$			27	8	27					
G56	$\overline{Q}7$			28	9	28					
G57	$\overline{Q}8$			29	11	29					
G58	$\overline{Q}9$			30	12	30					
G59	$\overline{Q}10$			31	13	31					
G60	$\overline{Q}11$			32	14	32					
G61	$\overline{Q}12$			33	15	33					
G62	$\overline{Q}13$			34	16	34					
G63	$\overline{Q}14$			35	17	35					
G64	$\overline{Q}15$			36	18	36					
G65	Q? Output voltage (B high level, A, C, and D low level)	38	37,39, 40	4	4	23					
G66	$\overline{Q}0$			21	2	21					
G67	$\overline{Q}1$			22	3	22					
G68	$\overline{Q}3$			24	5	24					
G69	$\overline{Q}4$			25	6	25					
[illegible]	[illegible]			26	7	26					

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Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
G71	$\overline{Q}6$ Output voltage (B high level, A, C, and D low level)	38	37,39, 40	27	8	27	19,20	0	400	mV	9
G72	$\overline{Q}7$			28	9	28					
G73	$\overline{Q}8$			29	11	29					
G74	$\overline{Q}9$			30	12	30					
G75	$\overline{Q}10$			31	13	31					
G76	$\overline{Q}11$			32	14	32					
G77	$\overline{Q}12$			33	15	33					
G78	$\overline{Q}13$			34	16	34					
G79	$\overline{Q}14$			35	17	35					
G80	$\overline{Q}15$			36	18	36					
G81	Q3 Output voltage (A and B high level, C and D low level)	37,38	39,40	5	5	24					
G82	$\overline{Q}0$			21	2	21					
G83	$\overline{Q}1$			22	3	22					
G84	$\overline{Q}2$			23	4	23					
G85	$\overline{Q}4$			25	6	25					
G86	$\overline{Q}5$			26	7	26					
G87	$\overline{Q}6$			27	8	27					
G88	$\overline{Q}7$			28	9	28					
G89	$\overline{Q}8$			29	11	29					
G90	$\overline{Q}9$			30	12	30					
G91	$\overline{Q}10$			31	13	31					
G92	$\overline{Q}_{1}$			32	14	32					
G93	$\overline{Q}12$			33	15	33					
G94	$\overline{Q}13$			34	16	34					
G95	$\overline{Q}_{1}$			35	17	35					
G96	$\overline{Q}$				18						

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* TABLE VII. Output voltage (low-level and high-level) (threshold inputs) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
G97	$\overline{Q}4$ Output voltage (C high level, A, B, and D low level)	39	37,38, 40	6	6	25	19,20	0	400	mV	9
G98	$\overline{Q}0$			21	2	21					
G99	$\overline{Q}1$			22	3	22					
G100	$\overline{Q}2$			23	4	23					
G101	$\overline{Q}3$			24	5	24					
G102	$\overline{Q}5$			26	7	26					
G103	$\overline{Q}6$			27	8	27					
G104	$\overline{Q}7$			28	9	28					
G105	$\overline{Q}8$			29	11	29					
G106	$\overline{Q}9$			30	12	30					
G107	$\overline{Q}10$			31	13	31					
G108	$\overline{Q}11$			32	14	32					
G109	$\overline{Q}12$			33	15	33					
G110	$\overline{Q}13$			34	16	34					
G111	$\overline{Q}14$			35	17	35					
G112	$\overline{Q}15$			36	18	36					
G113	Q5 Output voltage (A and C high level, B and D low level)	37,39	38,40	7	7	26					
G114	$\overline{Q}0$			21	2	21					
G115	$\overline{Q}1$			22	3	22					
G116	$\overline{Q}2$			23	4	23					
G117	$\overline{Q}3$			24	5	24					
G118	$\overline{Q}4$			25	6	25					

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* TABLE VII. Output voltage (low-level and high-level) (threshold inputs) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
G119	$\overline{Q}$6 Output voltage (A and C high level, B, and D low level)	37,39	38,40	27	8	27	19,20	0	400	mV	9
G120	$\overline{Q}$7			28	9	28					
G121	$\overline{Q}$8			29	11	29					
G122	$\overline{Q}$9			30	12	30					
G123	$\overline{Q}$10			31	13	31					
G124	$\overline{Q}$11			32	14	32					
G125	$\overline{Q}$12			33	15	33					
G126	$\overline{Q}$13			34	16	34					
G127	$\overline{Q}$14			35	17	35					
G128	$\overline{Q}$15			36	18	36					
G129	Q6 Output voltage (B and C high level, A and D low level)	38,39	37,40	8	8	27					
G130	$\overline{Q}$0			21	2	21					
G131	$\overline{Q}$1			22	3	22					
G132	$\overline{Q}$2			23	4	23					
G133	$\overline{Q}$3			24	5	24					
G134	$\overline{Q}$4			25	6	25					
G135	$\overline{Q}$5			26	7	26					
G136	$\overline{Q}$7			28	9	28					
G137	$\overline{Q}$8			29	11	29					
G138	$\overline{Q}$9			30	12	30					
G139	$\overline{Q}$10			31	13	31					
G140	$\overline{Q}$11			32	14	32					

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* TABLE VII. Output voltage (low-level and high-level) (threshold inputs) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
G141	$\overline{Q}12$ Output voltage (B and C high level, A and D low level)	38,39	37,40	33	15	33	19,20	0	400	mV	9
G142	$\overline{Q}13$			34	16	34					
G143	$\overline{Q}14$			35	17	35					
G144	$\overline{Q}15$			36	18	36					
G145	Q7 Output voltage (A, B, and C high level, D low level)	37,38, 39	40	9	9	28					
G146	$\overline{Q}0$			21	2	21					
G147	$\overline{Q}1$			22	3	22					
G148	$\overline{Q}2$			23	4	23					
G149	$\overline{Q}3$			24	5	24					
G150	$\overline{Q}4$			25	6	25					
G151	$\overline{Q}5$			26	7	26					
G152	$\overline{Q}6$			27	8	27					
G153	$\overline{Q}8$			29	11	29					
G154	$\overline{Q}9$			30	12	30					
G155	$\overline{Q}10$			31	13	31					
G156	$\overline{Q}11$			32	14	32					
G157	$\overline{Q}12$			33	15	33					
G158	$\overline{Q}13$			34	16	34					
G159	$\overline{Q}14$			35	17	35					
G160	$\overline{Q}15$			36	18	36					

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* TABLE VII. Output voltage (low-level and high-level) (threshold inputs) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
G161	Q8 Output voltage (D high level, A, B, and C low level)	40	37,38, 39	11	11	29	19,20	0	400	mV	9
G162	$\overline{Q}$0			21	2	21					
G163	$\overline{Q}$1			22	3	22					
G164	$\overline{Q}$2			23	4	23					
G165	$\overline{Q}$3			24	5	24					
G166	$\overline{Q}$4			25	6	25					
G167	$\overline{Q}$5			26	7	26					
G168	$\overline{Q}$6			27	8	27					
G169	$\overline{Q}$7			28	9	28					
G170	$\overline{Q}$9			30	12	30					
G171	$\overline{Q}$10			31	13	31					
G172	$\overline{Q}$11			32	14	32					
G173	$\overline{Q}$12			33	15	33					
G174	$\overline{Q}$13			34	16	34					
G175	$\overline{Q}$14			35	17	35					
G176	$\overline{Q}$15			36	18	36					
G177	Q9 Output voltage (A and D high level, B and C low level)	37,40	38,39,	12	12	30					
G178	$\overline{Q}$0			21	2	21					
G179	$\overline{Q}$1			22	3	22					
G180	$\overline{Q}$2			23	4	23					
G181	$\overline{Q}$3			24	5	24					
G182	$\overline{Q}$4			25	6	25					

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* TABLE VII. Output voltage (low-level and high-level) (threshold inputs) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
G183	$\overline{Q}5$ Output voltage (A and D high level, B and C low level)	37,40	38,39	26	7	26	19,20	0	400	mV	9
G184	$\overline{Q}6$			27	8	27					
G185	$\overline{Q}7$			28	9	28					
G186	$\overline{Q}8$			29	11	29					
G187	$\overline{Q}10$			31	13	31					
G188	$\overline{Q}11$			32	14	32					
G189	$\overline{Q}12$			33	15	33					
G190	$\overline{Q}13$			34	16	34					
G191	$\overline{Q}14$			35	17	35					
G192	$\overline{Q}15$			36	18	36					
G193	Q10 Output voltage (B and D high level, A and C low level)	38,40	37,39	13	13	31					
G194	$\overline{Q}0$			21	2	21					
G195	$\overline{Q}1$			22	3	22					
G196	$\overline{Q}2$			23	4	23					
G197	$\overline{Q}3$			24	5	24					
G198	$\overline{Q}4$			25	6	25					
G199	$\overline{Q}5$			26	7	26					
G200	$\overline{Q}6$			27	8	27					
G201	$\overline{Q}7$			28	9	28					
G202	$\overline{Q}8$			29	11	29					
G203	$\overline{Q}9$			30	12	30					

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* TABLE VII. Output voltage (low-level and high-level) (threshold inputs) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
G204	$\overline{Q}$11 Output voltage (B and D high level, A and C low level)	38,40	37,39	32	14	32	19,20	0	400	mV	9
G205	$\overline{Q}$12			33	15	33					
G206	$\overline{Q}$13			34	16	34					
G207	$\overline{Q}$14			35	17	35					
G208	$\overline{Q}$15			36	18	36					
G209	Q11 Output voltage (A, B, and D high level, C low level)	37,38, 40	39	14	14	32					
G210	$\overline{Q}$0			21	2	21					
G211	$\overline{Q}$1			22	3	22					
G212	$\overline{Q}$2			23	4	23					
G213	$\overline{Q}$3			24	5	24					
G214	$\overline{Q}$4			25	6	25					
G215	$\overline{Q}$5			26	7	26					
G216	$\overline{Q}$6			27	8	27					
G217	$\overline{Q}$7			28	9	28					
G218	$\overline{Q}$8			29	11	29					
G219	$\overline{Q}$9			30	12	30					
G220	$\overline{Q}$10			31	13	31					
G221	$\overline{Q}$12			33	15	33					
G222	$\overline{Q}$13			34	16	34					
G223	$\overline{Q}$14			35	17	35					
G224	$\overline{Q}$15			36	18	36					

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* TABLE VII. Output voltage (low-level and high-level) (threshold inputs) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
G225	Q12 Output voltage (C and D high level, A and B low level)	39,40	37,38	15	15	33	19,20	0	400	mV	9
G226	$\overline{Q}0$			21	2	21					
G227	$\overline{Q}1$			22	3	22					
G228	$\overline{Q}2$			23	4	23					
G229	$\overline{Q}3$			24	5	24					
G230	$\overline{Q}4$			25	6	25					
G231	$\overline{Q}5$			26	7	26					
G232	$\overline{Q}6$			27	8	27					
G233	$\overline{Q}7$			28	9	28					
G234	$\overline{Q}8$			29	11	29					
G235	$\overline{Q}9$			30	12	30					
G236	$\overline{Q}10$			31	13	31					
G237	$\overline{Q}11$			32	14	32					
G238	$\overline{Q}13$			34	16	34					
G239	$\overline{Q}14$			35	17	35					
G240	$\overline{Q}15$			36	18	36					
G241	Q13 Output voltage (A, C, and D high level, B low level)	37,39, 40	38	16	16	34					
G242	$\overline{Q}0$			21	2	21					
G243	$\overline{Q}1$			22	3	22					
G244	$\overline{Q}2$			23	4	23					
G245	$\overline{Q}3$			24	5	24					
G246	$\overline{Q}4$			25	6	25					

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TABLE VII. Output voltage (low-level and high-level) (threshold inputs) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
G247	$\overline{Q}5$ Output voltage (A, C, and D high level, B low level)	37,39, 40	38	26	7	26	19,20	0	400	mV	9
G248	$\overline{Q}6$			27	8	27					
G249	$\overline{Q}7$			28	9	28					
G250	$\overline{Q}8$			29	11	29					
G251	$\overline{Q}9$			30	12	30					
G252	$\overline{Q}10$			31	13	31					
G253	$\overline{Q}11$			32	14	32					
G254	$\overline{Q}12$			33	15	33					
G255	$\overline{Q}14$			35	17	35					
G256	$\overline{Q}15$			36	18	36					
G257	Q14 Output voltage (B, C, and D high level, A low level)	38,39, 40	37	17	17	35					
G258	$\overline{Q}0$			21	2	21					
G259	$\overline{Q}1$			22	3	22					
G260	$\overline{Q}2$			23	4	23					
G261	$\overline{Q}3$			24	5	24					
G262	$\overline{Q}4$			25	6	25					
G263	Q5			26	7	26					
G264	$\overline{Q}6$			27	8	27					
G265	$\overline{Q}7$			28	9	28					
G266	$\overline{Q}8$			29	11	29					
G267	$\overline{Q}9$			30	12	30					
G268	$\overline{Q}10$			31	13	31					
G269	$\overline{Q}11$			32	14	32					
G270	$\overline{Q}12$			33	15	33					
G271	$\overline{Q}13$			34	16	34					
G272	$\overline{Q}15$			36	18	36					

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* TABLE VII Output voltage (low-level and high-level) (threshold inputs) test - Continued

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
G273	$\overline{Q}15$ Output voltage (A, B, C, and D high level)	37,38, 39,40	Open	18	18	36	19,20	0	400	mV	9
G274	$\overline{Q}0$			21	2	21					
G275	$\overline{Q}1$			22	3	22					
G276	$\overline{Q}2$			23	4	23					
G277	$\overline{Q}3$			24	5	24					
G278	$\overline{Q}4$			25	6	25					
G279	$\overline{Q}5$			26	7	26					
G280	$\overline{Q}6$			27	8	27					
G281	$\overline{Q}7$			28	9	28					
G282	$\overline{Q}8$			29	11	29					
G283	$\overline{Q}9$			30	12	30					
G284	$\overline{Q}10$			31	13	31					
G285	$\overline{Q}11$			32	14	32					
G286	$\overline{Q}12$			33	15	33					
G287	$\overline{Q}13$			34	16	34					
G288	$\overline{Q}14$			35	17	35					
H1	$\overline{Q}0$ Output voltage (enable low level)	Open	19,20	21	2	21	37,38,39,40	2.4	4 5	V	
H2	$\overline{Q}1$			22	3	22	38,39,40				
H3	$\overline{Q}2$			23	4	23	37,39,40				
H4	$\overline{Q}3$			24	5	24	39,40				
H5	$\overline{Q}4$			25	6	25	37,38,40				
H6	$\overline{Q}5$			26	7	26	38,40				
H7	$\overline{Q}6$			27	8	27	37,40				
H8	$\overline{Q}7$			28	9	28	40				

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* TABLE VII. Output voltage (low-level and high-level) (threshold inputs) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
H9	$\overline{Q}$8 Output voltage (enable low level)	Open	19,20	29	11	29	37,39	2.4	4.5	V	9
H10	$\overline{Q}$9			30	12	30	38,39				
H11	$\overline{Q}$10			31	13	31	37,39				
H12	$\overline{Q}$11			32	14	32	39				
H13	$\overline{Q}$12			33	15	33	37,38				
H14	$\overline{Q}$13			34	16	34	38				
H15	$\overline{Q}$14			35	17	35	37				
H16	$\overline{Q}$15			36	18	36	Open				
H17	Q0 Output voltage (enable high level)	20	19	2	2	21	38,39,40				
H18	Q1			3	3	22	37,38,39,40				
H19	Q2			4	4	23					
H20	Q3			5	5	24					
H21	Q4			6	6	25					
H22	Q5			7	7	26					
H23	Q6			8	8	27					
H24	Q7			9	9	28					
H25	Q8			11	11	29					
H26	Q9			12	12	30					
H27	Q10			13	13	31					
H28	Q11			14	14	32					
H29	Q12			15	15	33					
H30	Q13			16	16	34					
H31	Q14			17	17	35					
H32	Q15			18	18	36					

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* TABLE VII Output voltage (low-level and high-level) (threshold inputs) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
H33	$\overline{Q0}$ Output voltage (A, B, C, and D low level)	Open	37,38, 39,40	21	2	21	19,20	2.4	4 5	V	9
H34	Q1			3	3	22					
H35	Q2			4	4	23					
H36	Q3			5	5	24					
H37	Q4			6	6	25					
H38	Q5			7	7	26					
H39	Q6			8	8	27					
H40	Q7			9	9	28					
H41	Q8			11	11	29					
H42	Q9			12	12	30					
H43	Q10			13	13	31					
H44	Q11			14	14	32					
H45	Q12			15	15	33					
H46	Q13			16	16	34					
H47	Q14			17	17	35					
H48	Q15			18	18	36					
H49	$\overline{Q1}$ Output voltage (A high level, B, C, and D low level)	37	38,39, 40	22	3	22					
H50	Q0			2	2	21					
H51	Q2			4	4	23					
H52	Q3			5	5	24					
H53	Q4			6	6	25					
H54	Q5			7	7	26					

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* TABLE VII. Output voltage (low-level and high-level) (threshold inputs) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
H55	Q6 Output voltage (A high level, B, C, and D low level)	37	38,39, 40	8	8	27	19,20	2.4	4.5	V	9
H56	Q7			9	9	28					
H57	Q8			11	11	29					
H58	Q9			12	12	30					
H59	Q10			13	13	31					
H60	Q11			14	14	32					
H61	Q12			15	15	33					
H62	Q13			16	16	34					
H63	Q14			17	17	35					
H64	Q15			18	18	36					
H65	$\overline{Q2}$ Output voltage (B high level, A, C, and D low level)	38	37,39, 40	23	4	23					
H66	Q0			2	2	21					
H67	Q1			3	3	22					
H68	Q3			5	5	24					
H69	Q4			6	6	25					
H70	Q5			7	7	26					
H71	Q6			8	8	27					
H72	Q7			9	9	28					
H73	Q8			11	11	29					
H74	Q9			12	12	30					
H75	Q10			13	13	31					

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* TABLE VII. Output voltage (low level and high-level) (threshold inputs) test - Continued.

Test number	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
H76	Q11 Output voltage (B high level, A, C, and D low level)	38	37,39, 40	14	14	32	19,20	2.4	4.5	V	9
H77	Q12			15	15	33					
H78	Q13			16	16	34					
H79	Q14			17	17	35					
H80	Q15			18	18	36					
H81	Q3 Output voltage (A and B high level, C and D low level)	37,38	39,40	24	5	24					
H82	Q0			2	2	21					
H83	Q1			3	3	22					
H84	Q2			4	4	23					
H85	Q4			6	6	25					
H86	Q5			7	7	26					
H87	Q6			8	8	27					
H88	Q7			9	9	28					
H89	Q8			11	11	29					
H90	Q9			12	12	30					
H91	Q10			13	13	31					
H92	Q11			14	14	32					
H93	Q12			15	15	33					
H94	Q13			16	16	34					
H95	Q14			17	17	35					
H96	Q15			18	18	36					

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* TABLE VII. Output voltage (low-level and high-level) (threshold inputs) test - Continued.

Test condition		Parameter	Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
H97	Q4	Output voltage (C high level, A, B, and D low level)	39	37,38, 40	25	6	25	19,20	2.4	4.5	V	9
H98	Q0				2	2	21					
H99	Q1				3	3	22					
H100	Q2				4	4	23					
H101	Q3				5	5	24					
H102	Q5				7	7	26					
H103	Q6				8	8	27					
H104	Q7				9	9	28					
H105	Q8				11	11	29					
H106	Q9				12	12	30					
H107	Q10				13	13	31					
H108	Q11				14	14	32					
H109	Q12				15	15	33					
H110	Q13				16	16	34					
H111	Q14				17	17	35					
H112	Q15				18	18	36					
H113	Q5	Output voltage (A and C high level, B and D low level)	37,39	38,40	26	7	26					
H114	Q0				2	2	21					
H115	Q1				3	3	22					
H116	Q2				4	4	23					
H117	Q3				5	5	24					
H118	Q4				6	6	25					

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* TABLE VII. Output voltage (low-level and high-level) (threshold inputs) test - Continued.

Test number	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
H119	Q6 Output voltage (A and C high level, B and D low level)	37,39	38,40	8	8	27	19,20	2.4	4.5	V	9
H120	Q7			9	9	28					
H121	Q8			11	11	29					
H122	Q9			12	12	30					
H123	Q10			13	13	31					
H124	Q11			14	14	32					
H125	Q12			15	15	33					
H126	Q13			16	16	34					
H127	Q14			17	17	35					
H128	Q15			18	18	36					
H129	Q6 Output voltage (B and C high level, A and D low level)	38,39	37,40	27	8	27					
H130	Q0			2	2	21					
H131	Q1			3	3	22					
H132	Q2			4	4	23					
H133	Q3			5	5	24					
H134	Q4			6	6	25					
H135	Q5			7	7	26					
H136	Q7			9	9	28					
H137	Q8			11	11	29					
H138	Q9			12	12	30					
H139	Q10			13	13	31					
H140	Q11			14	14	32					

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* TABLE VII. Output voltage (low-level and high-level) (threshold inputs) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
H141	Q12 Output voltage (B and C high level, A and D low level)	38,39	37,40	15	15	33	19,20	2.4	4.5	V	9
H142	Q13			16	16	34					
H143	Q14			17	17	35					
H144	Q15			18	18	36					
H145	$\bar{Q}$7 Output voltage (A, B, and C high level, D low level)	37,38, 39	40	28	9	28					
H146	Q0			2	2	21					
H147	Q1			3	3	22					
H148	Q2			4	4	23					
H149	Q3			5	5	24					
H150	Q4			6	6	25					
H151	Q5			7	7	26					
H152	Q6			8	8	27					
H153	Q8			11	11	29					
H154	Q9			12	12	30					
H155	Q10			13	13	31					
H156	Q11			14	14	32					
H157	Q12			15	15	33					
H158	Q13			16	16	34					
H159	Q14			17	17	35					
H160	Q15			18	18	36					

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* TABLE VII. Output voltage (low-level and high-level) (threshold inputs) test - Continued.

Test condition		Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
F..	Q6	Output voltage (D high leve, A, B, and C low level)	40	37,38, 39	29	11	29	19,20	2.4	4.5	V	9
H162	Q0				2	2	21					
H163	Q1				3	3	22					
H164	Q2				4	4	23					
H165	Q3				5	5	24					
H166	Q4				6	6	25					
H167	Q5				7	7	26					
H168	Q6				8	8	27					
H169	Q7				9	9	28					
H170	Q9				12	12	30					
H171	Q10				13	13	31					
H17.	Q11				14	14	32					
H173	Q12				15	15	33					
H174	Q13				16	16	34					
H175	Q14				17	17	35					
H176	Q15				18	18	36					
H17.	Q9	Output voltage (A and D high level, B and C low level)	37,40	38,39,	30	12	30					
H178	Q0				2	2	21					
H179	Q1				3	3	22					
H180	Q2				4	4	23					
H181	Q3				5	5	24					
H182	Q4				6	6	25					

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* TABLE VII. Output voltage (low-level and high-level) (threshold inputs) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
H183	Q5 Output voltage (A and D high level, B and C low level)	37,40	38,39	7	7	26	19,20	2.4	4.5	V	9
H184	Q6			8	8	27					
H185	Q7			9	9	28					
H186	Q8			11	11	29					
H187	Q10			13	13	31					
H188	Q11			14	14	32					
H189	Q12			15	15	33					
H190	Q13			16	16	34					
H191	Q14			17	17	35					
H192	Q15			18	18	36					
H193	Q10 Output voltage (B and D high level, A and C low level)	38,40	37,39	31	13	31					
H194	Q0			2	2	21					
H195	Q1			3	3	22					
H196	Q2			4	4	23					
H197	Q3			5	5	24					
H198	Q4			6	6	25					
H199	Q5			7	7	26					
H200	Q6			8	8	27					
H201	Q7			9	9	28					
H202	Q8			11	11	29					
H203	Q9			12	12	30					

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* TABLE VII. Output voltage (low-level and high-level) (threshold inputs) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
H204	Q11 Output voltage (B and D high level, A and C low level)	38,40	37,39	14	14	32	19,20	2.4	4.5	V	9
H205	Q12			15	15	33					
H206	Q13			16	16	34					
H207	Q14			17	17	35					
H208	Q15			18	18	36					
H209	Q11 Output voltage (A, B, and D high level, C low level)	37,38, 40	39	32	14	32					
H210	Q0			2	2	21					
H211	Q1			3	3	22					
H212	Q2			4	4	23					
H213	Q3			5	5	24					
H214	Q4			6	6	25					
H215	Q5			7	7	26					
H216	Q6			8	8	27					
H217	Q7			9	9	28					
H218	Q8			11	11	29					
H219	Q9			12	12	30					
H220	Q10			13	13	31					
H221	Q12			15	15	33					
H222	Q13			16	16	34					
H223	Q14			17	17	35					
H224	Q15			18	18	36					

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* TABLE VII. Output voltage (low-level and high-level) (threshold inputs) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
H225	$\overline{Q}$12 Output voltage (C and D high level, A and B low level)	39,40	37,38	33	15	33	19,20	2.4	4.5	V	9
H226	Q0			2	2	21					
H227	Q1			3	3	22					
H228	Q2			4	4	23					
H229	Q3			5	5	24					
H230	Q4			6	6	25					
H231	Q5			7	7	26					
H232	Q6			8	8	27					
H233	Q7			9	9	28					
H234	Q8			11	11	29					
H235	Q9			12	12	30					
H236	Q10			13	13	31					
H237	Q11			14	14	32					
H238	Q13			16	16	34					
H239	Q14			17	17	35					
H240	Q15			18	18	36					
H241	$\overline{Q}$13 Output voltage (A, C, and D high level, B low level)	37,39, 40	38	34	16	34					
H242	Q0			2	2	21					
H243	Q1			3	3	22					
H244	Q2			4	4	23					
H245	Q3			5	5	24					
H246	Q4			6	6	25					

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* TABLE VII. Output voltage (low-level and high-level) (threshold inputs) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
H247	Q5 Output voltage (A, C, and D high level, B low level)	37,39, 40	38	7	7	26	19,20	2.4	4.5	V	9
H248	Q6			8	8	27					
H249	Q7			9	9	28					
H250	Q8			11	11	29					
H251	Q9			12	12	30					
H252	Q10			13	13	31					
H253	Q11			14	14	32					
H254	Q12			15	15	33					
H255	Q14			17	17	35					
H256	Q15			18	18	36					
H257	Q14 Output voltage (B, C, and D high level, A low level)	38,39, 40	37	35	17	35					
H258	Q0			2	2	21					
H259	Q1			3	3	22					
H260	Q2			4	4	23					
H261	Q3			5	5	24					
H262	Q4			6	6	25					
H263	Q5			7	7	26					
H264	Q6			8	8	27					
H265	Q7			9	9	28					
H266	Q8			11	11	29					
H267	Q9			12	12	30					
H268	Q10			13	13	31					
H269	Q11			14	14	32					
H270	Q12			15	15	33					
H271	Q13			16	16	34					
				18	18	36					

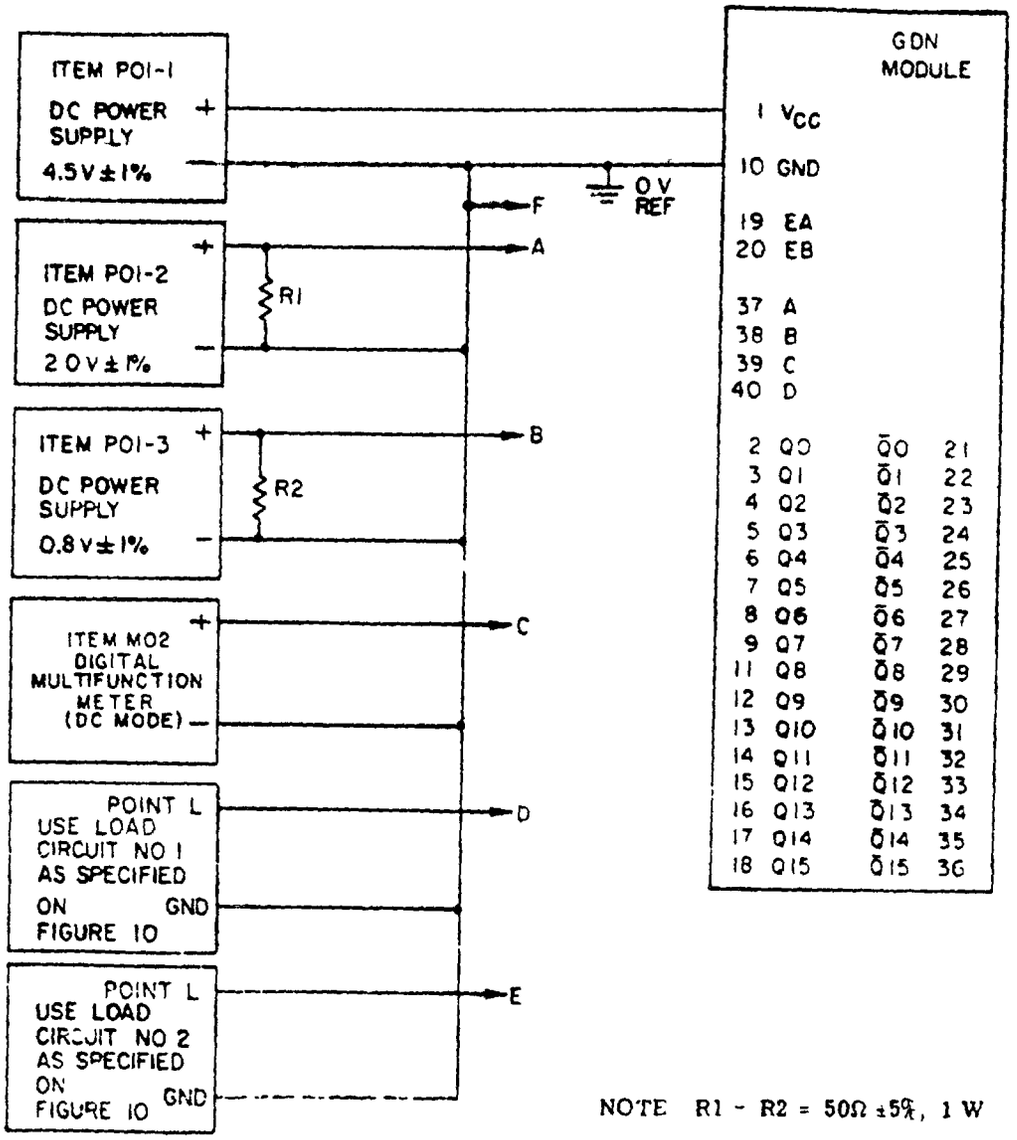
MIL-M-38510/75(EC)

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* TABLE VII. Output voltage (low-level and high-level) (threshold inputs) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
H273	Q15 Output voltage (A, B, C, and D high level)	37,38, 39,40	Open	36	18	36	19,20	2.4	4.5	V	9
H274	Q0			2	2	21					
H275	Q1			3	3	22					
H276	Q2			4	4	23					
H277	Q3			5	5	24					
H278	Q4			6	6	25					
H279	Q5			7	7	26					
H280	Q6			8	8	27					
H281	Q7			9	9	28					
H282	Q8			11	11	29					
H283	Q9			12	12	30					
H284	Q10			13	13	31					
H285	Q11			14	14	32					
H286	Q12			15	15	33					
H287	Q13			16	16	34					
H288	Q14			17	17	35					

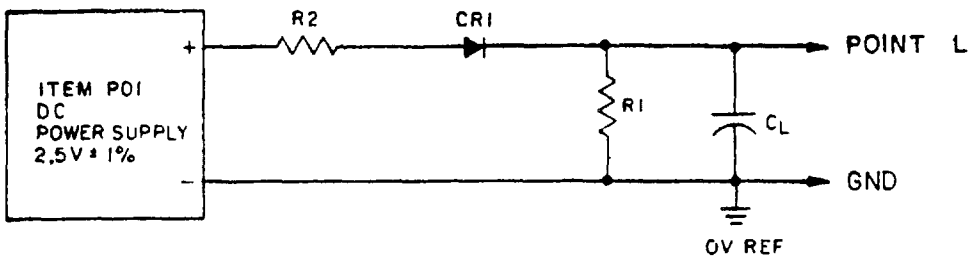
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NOTE R1 - R2 = 50Ω ±5%, 1 W

* FIGURE 9. Output voltage (low-level and high-level) (threshold inputs) test circuit for table VII

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NOTES

1. Load circuit No. 1 (Q outputs)

   (a) CR1 = 1N4150, or equivalent.
   (b) C_L = 130 ±5 pF (including probe and parasitic capacitance).
   (c) For test conditions R and S, C_L = 50 ±5 pF (including probe and parasitic capacitance).
   (d) R1 = 3.16 kΩ ±1%, 1/8 W.
   (e) R2 = Select value for a current flow of 15.6 mA ±1% from point L to ground

2. Load circuit No. 2 ($\overline{Q}$ outputs)

   (a) CR1 = 1N4150, or equivalent.
   (b) C_L = 130 ±5 pF (including probe and parasitic capacitance).
   (c) For test conditions R and S, C_L = 50 ±5 pF (including probe and parasitic capacitance)
   (d) R1 = 6.04 kΩ ±1%, 1/8 W
   (e) R2 = Select value for a current flow of 17.3 mA ±1% from point L to ground.

3. Capacitor C_L is not required for dc electrical conditions G or H.
4. During output switching, the + 5 V power supply shall have a maximum overshoot of 500 mV peak.

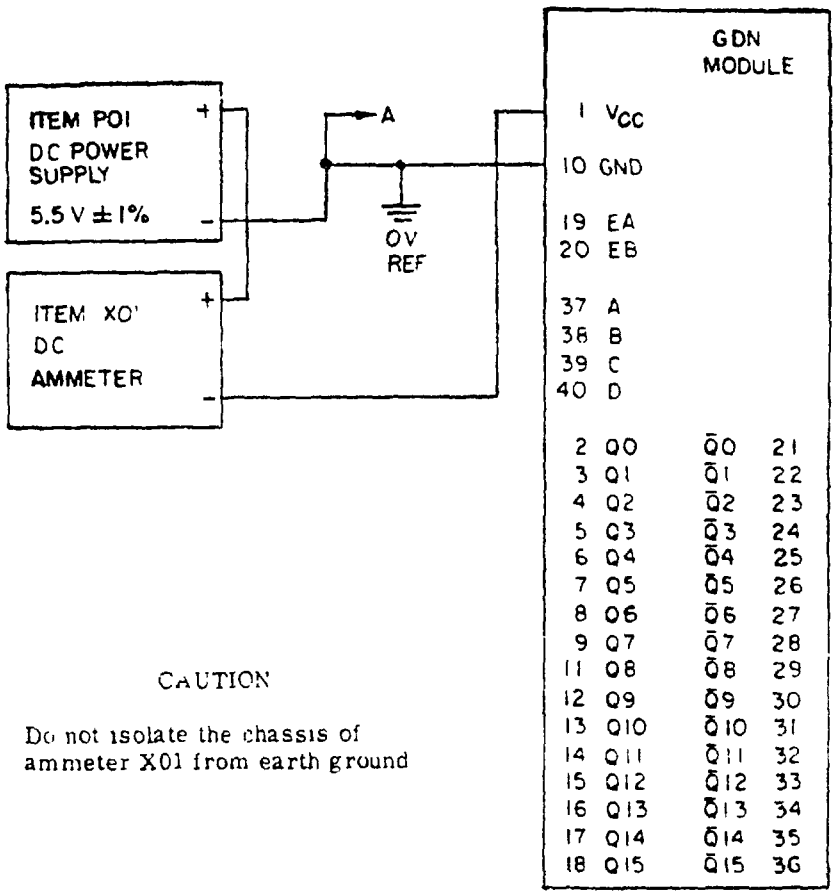
FIGURE 10. GDN logic load circuit

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* TABLE VIII. Power supply current.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
J	Power supply current	19,20,37,38,39,40	80	210	mA	11

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* FIGURE 11 Power supply current test circuit for table VII.

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* TABLE IX Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
K1	Enable to QD Transition time (t_{TLH}), C_L = 130 pF	19	2	2	21	Open	20,37,38,39,40	5	50	ns	12,13
K2	Enable to $\overline{Q0}$		21	2	21	Open	20,37,38,39,40				
K3	Enable to Q1		3	3	22	37	20,38,39,40				
K4	Enable to $\overline{Q1}$		22	3	22	37	20,38,39,40				
K5	Enable to Q2		4	4	23	38	20,37,39,40				
K6	Enable to $\overline{Q2}$		23	4	23	38	20,37,39,40				
K7	Enable to Q3		5	5	24	37,38	20,39,40				
K8	Enable to $\overline{Q3}$		24	5	24	37,38	20,39,40				
K9	Enable to Q4		6	6	25	39	20,37,38,40				
K10	Enable to $\overline{Q4}$		25	6	25	39	20,37,38,40				
K11	Enable to Q5		7	7	26	37,39	20,38,40				
K12	Enable to $\overline{Q5}$		26	7	26	37,39	20,38,40				
K13	Enable to Q6		8	8	27	38,39	20,37,40				
K14	Enable to $\overline{Q6}$		27	8	27	38,39	20,37,40				

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* TABLE IX. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
K15	Enable to Q7 Transition time (t_{TLH}), C_L = 130 pF	19	9	9	28	37,38, 39	20,40	5	50	ns	12,13
K16	Enable to Q7		28	9	28	37,38, 39	20,40				
K17	Enable to Q8		11	11	29	40	20,37,38,39				
K18	Enable to Q8		29	11	29	40	20,37,38,39				
K19	Enable to Q9		12	12	30	37,40	20,38,39				
K20	Enable to Q9		30	12	30	37,40	20,38,39				
K21	Enable to Q10		13	13	31	38,40	20,37,39				
K22	Enable to Q10		31	13	31	38,40	20,37,39				
K23	Enable to Q11		14	14	32	37,38, 40	20,39				
K24	Enable to Q11		32	14	32	37,38, 40	20,39				
K25	Enable to Q12		15	15	33	39,40	20,37,38				
K26	Enable to Q12		33	15	33	39,40	20,37,38				
K27	Enable to Q13		16	16	34	37,39, 40	20,38				
K28	Enable to Q13		34	16	34	37,39, 40	20,38				

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* TABLE IX. Transition time (t_{TLH} and t_{THL}) test - Continued

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
K29	Enable to Q14 Transition time (t_{TLH}), C_L = 130 pF	19	17	17	35	38,39, 40	20,37	5	50	ns	12,13
K30	Enable to $\overline{Q}$14		35	17	35	38,39, 40	20,37				
K31	Enable to Q15		18	18	36	37,38, 39,40	20				
K32	Enable to $\overline{Q}$15		36	18	36	37,38, 39,40	20				
L1	Enable to Q0 Transition time (t_{THL}), C_L = 130 pF		2	2	21	Open	20,37,38,39,40		25		
L2	Enable to $\overline{Q}$0		21	2	21	Open	20,37,38,39,40				
L3	Enable to Q1		3	3	22	37	20,38,39,40				
L4	Enable to $\overline{Q}$1		22	3	22	37	20,38,39,40				
L5	Enable to Q2		4	4	23	38	20,37,39,40				
L6	Enable to $\overline{Q}$2		23	4	23	38	20,37,39,40				
L7	Enable to Q3		5	5	24	37,38	20,39,40				
L8	Enable to $\overline{Q}$3		24	5	24	37,38	20,39,40				
L9	Enable to Q4		6	6	25	39	20,37,38,40				
L10	Enable to $\overline{Q}$4		25	6	25	39	20,37,38,40				
L11	Enable to Q5		7	7	26	37,39	20,38,40				
L12	Enable to $\overline{Q}$5		26	7	26	37,39	20,38,40				
L13	Enable to Q6		8	8	27	38,39	20,37,40				
L14	Enable to $\overline{Q}$6		27	8	27	38,39	20,37,40				

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* TABLE IX. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
L15	Enable to Q7 Transition time (t_{THL}). C_L = 130 pF	19	9	9	28	37,38, 39	20,40	6	25	ns	12,13
L16	Enable to $\overline{Q7}$		28	9	28	37,38, 39	20,40				
L17	Enable to Q8		11	11	29	40	20,37,38,39				
L18	Enable to $\overline{Q8}$		29	11	29	40	20,37,38,39				
L19	Enable to Q9		12	12	30	37,40	20,38,39				
L20	Enable to $\overline{Q9}$		30	12	30	37,40	20,38,39				
L21	Enable to Q10		13	13	31	38,40	20,37,39				
L22	Enable to $\overline{Q10}$		31	13	31	38,40	20,37,39				
L23	Enable to Q11		14	14	32	37,38, 40	20,39				
L24	Enable to $\overline{Q11}$		32	14	32	37,38, 40	20,39				
L25	Enable to Q12		15	15	33	39,40	20,37,38				
L26	Enable to $\overline{Q12}$		33	15	33	39,40	20,37,38				
L27	Enable to Q13		16	16	34	37,39, 40	20,38				
L28	Enable to $\overline{Q13}$		34	16	34	37,39, 40	20,38				

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* TABLE IX. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
L25	Enable to Q14	Transition time (t_{THL}), C_L = 130 pF	19	17	17	35	38,39, 40	20,37	5	25	ns	12,13
L30	Enable to $\overline{Q14}$			33	17	35	38,39, 40	20,37				
L31	Enable to Q15			18	18	36	37,38, 39,40	20				
L32	Enable to $\overline{Q15}$			33	18	36	37,38, 39,40	20				

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* TABLE X Propagation delay time (t_{PHL} and t_{PLH}) test.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
M1	Enable to Q0 Propagation delay time (t_{PHL}), C_L = 130 pF	19	2	2	21	Open	20,37,38,39,40	5	32	ns	12,13
M2	Enable to $\overline{Q0}$		21	2	21	Open	20,37,38,39,40		49		
M3	Enable to Q1		3	3	22	37	20,36,39,40		32		
M4	Enable to $\overline{Q1}$		22	3	22	37	20,38,39,40		49		
M5	Enable to Q2		4	4	23	38	20,37,39,40		32		
M6	Enable to $\overline{Q2}$		23	4	23	38	20,37,39,40		49		
M7	Enable to Q3		5	5	24	37,38	20,39,40		32		
M8	Enable to $\overline{Q3}$		24	5	24	37,38	20,39,40		49		
M9	Enable to Q4		6	6	25	39	20,37,38,40		32		
M10	Enable to $\overline{Q4}$		25	6	25	39	20,37,38,40		49		
M11	Enable to Q5		7	7	26	37,39	20,38,40		32		
M12	Enable to $\overline{Q5}$		26	7	26	37,39	20,38,40		49		
M13	Enable to Q6		8	8	27	38,39	20,37,40		32		
M14	Enable to $\overline{Q6}$		27	8	27	38,39	20,37,40		49		

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* TABLE X. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
M15	Enable to Q7 Propagation delay time (t_{PHL}), C_L = 130 pF	19	9	9	28	37,38, 39	20,40	5	32	ns	12,13
M16	Enable to $\overline{Q7}$		28	9	28	37,38, 39	20,40		49		
M17	Enable to Q8		11	11	29	40	20,37,38,39		32		
M18	Enable to $\overline{Q8}$		29	11	29	40	20,37,38,39		49		
M19	Enable to Q9		12	12	30	37,40	20,38,39		32		
M20	Enable to $\overline{Q9}$		30	12	30	37,40	20,38,39		49		
M21	Enable to Q10		13	13	31	38,40	20,37,39		32		
M22	Enable to $\overline{Q10}$		31	13	31	38,40	20,37,39		49		
M23	Enable to Q11		14	14	32	37,38, 40	20,39		32		
M24	Enable to $\overline{Q11}$		32	14	32	37,38, 40	20,39		49		
M25	Enable to Q12		15	15	33	39,40	20,37,38		32		
M26	Enable to $\overline{Q12}$		33	15	33	39,40	20,37,38		49		
M27	Enable to Q13		16	16	34	37,39, 40	20,38		32		
M28	Enable to $\overline{Q13}$		34	16	34	37,39, 40	20,38		49		

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* TABLE X. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
M29	Enable to Q14	Propagation delay time (t_{PHL}), C_L = 130 pF	19	17	17	35	38,39, 40	20,37	5	32	ns	12,13
M30	Enable to $\overline{Q14}$			35	17	35	38,39, 40	20,37		49		
M31	Enable to Q15			18	18	36	37,38, 39,40	20		32		
M32	Enable to $\overline{Q15}$			36	18	36	37,38, 39,40	20		49		
M33	A to Q0		37	2	2	21	Open	19,20,38,39,40		38		
M34	A to Q1		37	3	3	22		19,20,38,39,40				
M35	B to Q0		38	2	2	21		19,20,37,39,40				
M36	B to Q2		38	4	4	23		19,20,37,39,40				
M37	C to Q0		39	2	2	21		19,20,37,38,40				
M38	C to Q4		39	6	6	25		19,20,37,38,40				
M39	D to Q0		40	2	2	21		19,20,37,38,39				
M40	D to Q8		40	11	11	29		19,20,37,38,39				
N1	Enable to Q0	Propagation delay time (t_{PLH}), C_L = 130 pF	19	2	2	21	Open	20,37,38,39,40		32		
N2	Enable to $\overline{Q0}$			21	2	21	Open	20,37,38,39,40		65		
N3	Enable to Q1			3	3	22	37	20,38,39,40		32		
N4	Enable to $\overline{Q1}$			22	3	22	37	20,38,39,40		65		
N5	Enable to Q2			4	4	23	38	20,37,39,40		32		
N6	Enable to $\overline{Q2}$			23	4	23	38	20,37,39,40		65		

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* TABLE X. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Connect F to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
N7	Enable to Q3	19	5	5	24	37,38	20,39,40	5	32	ns	12,13
N8	Enable to $\overline{Q3}$		24	5	24	37,38	20,39,40		65		
N9	Enable to Q4		6	6	25	39	20,37,38,40		32		
N10	Enable to $\overline{Q4}$		25	6	25	39	20,37,38,40		65		
N11	Enable to Q5		7	7	26	37,39	20,38,40		32		
N12	Enable to $\overline{Q5}$		26	7	26	37,39	20,38,40		65		
N13	Enable to Q6		8	8	27	38,39	20,37,40		32		
N14	Enable to $\overline{Q6}$		27	8	27	38,39	20,37,40		65		
N15	Enable to Q7		9	9	28	37,38, 39	20,40		32		
N16	Enable to $\overline{Q7}$		28	9	28	37,38, 39	20,40		65		
N17	Enable to Q8		11	11	29	40	20,37,38,39		32		
N18	Enable to $\overline{Q8}$		29	11	29	40	20,37,38,39		65		
N19	Enable to Q9		12	12	30	37,40	20,38,39		32		
N20	Enable to $\overline{Q9}$		30	12	30	37,40	20,38,39		65		
N21	Enable to Q10		13	13	31	38,40	20,37,39		32		
N22	Enable to $\overline{Q10}$		31	13	31	38,40	20,37,39		65		

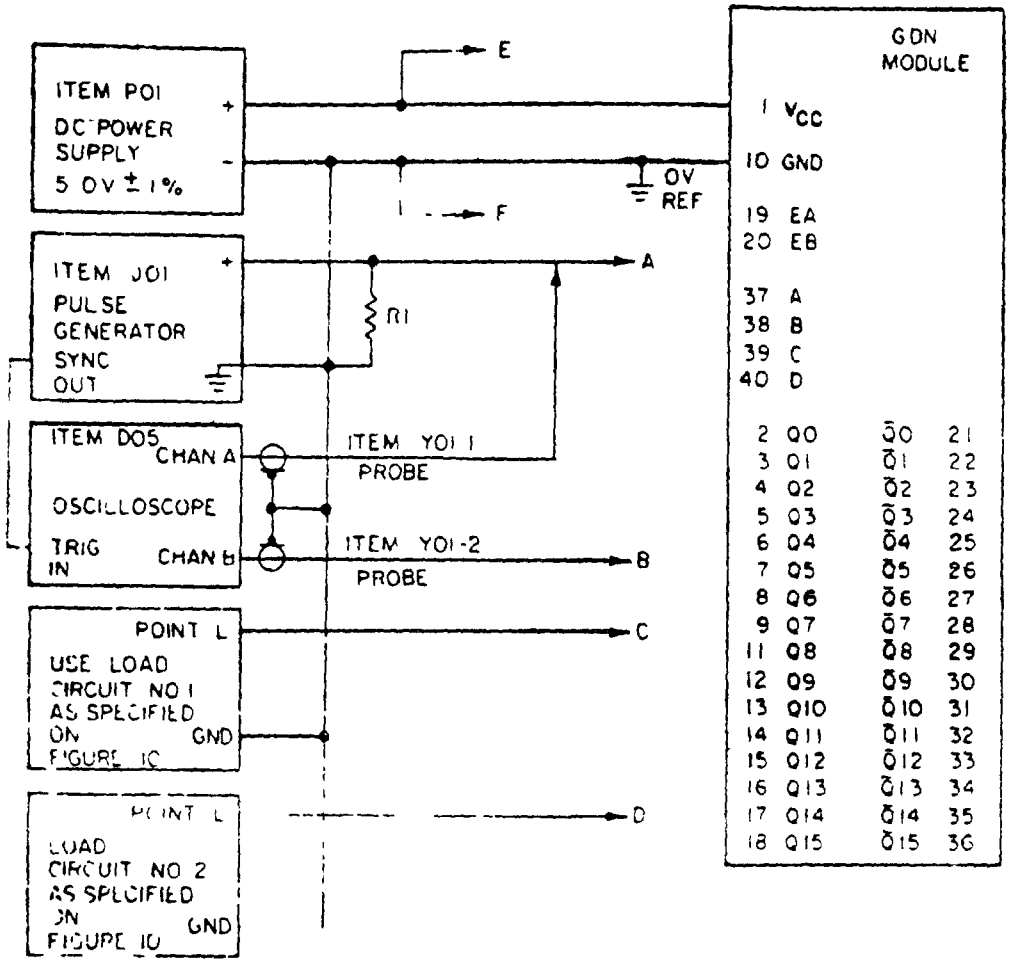
Propagation delay time (t_{PLH}), C_L = 130 pF

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* TABLE X. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
N23	Enable to Q11 Propagation delay time (t_{PLH}), C_L = 130 pF	19	14	14	32	37,38, 40	20,39	5	32	ns	12,13
N24	Enable to $\overline{Q11}$		32	14	32	37,38, 40	20,39		65		
N25	Enable to Q12		15	15	33	39,40	20,37,38		32		
N26	Enable to $\overline{Q12}$		33	15	33	39,40	20,37,38		65		
N27	Enable to Q13		16	16	34	37,39, 40	20,38		32		
N28	Enable to $\overline{Q13}$		34	16	34	37,39, 40	20,38		65		
N29	Enable to Q14		17	17	35	38,39, 40	20,37		32		
N30	Enable to $\overline{Q14}$		35	17	35	38,39, 40	20,37		65		
N31	Enable to Q15		18	18	36	37,38, 39,40	20		32		
N32	Enable to $\overline{Q15}$		36	18	36	37,38, 39,40	20		65		
N33	A to Q0	37	2	2	21	Open	19,20,38,39,40		38		
N34	A to Q1	37	3	3	22		19,20,38,39,40				
N35	B to Q0	38	2	2	21		19,20,37,39,40				
N36	B to Q2	39	4	4	23		19,20,37,39,40				
N37	C to Q0	39	2	2	21		19,20,37,38,40				
N38	C to Q4	39	6	6	25		19,20,37,38,40				
N39	D to Q0	40	2	2	21		19,20,37,38,39				
N40	D to Q8	40	11	11	29		19,20,37,38,39				

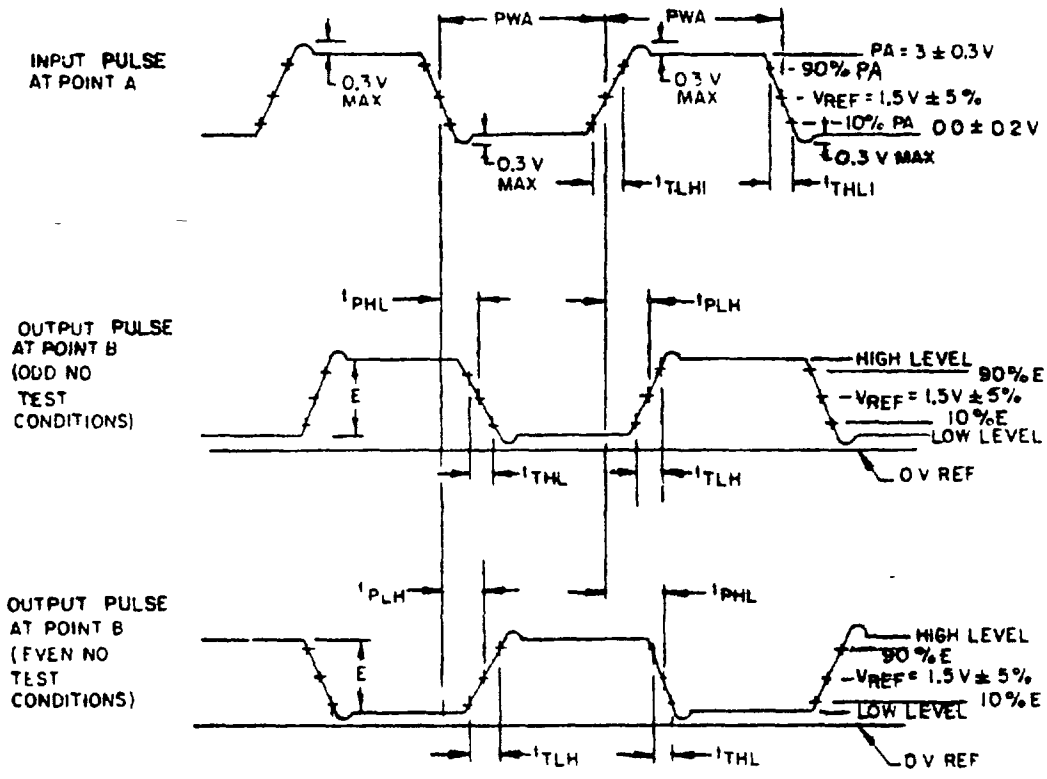
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NOTE R1 = 50Ω ±5%, 1 W

* FIGURE 12. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit for tables IX and X.

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NOTES
1. PWA = 500 ±50 ns
2. $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns
3. Before making the output measurements, the output levels shall be verified to be 0 4 V maximum for low level and 2 4 V minimum for high level
4. All voltage levels are referenced to ground
5. The lead length of electrical conductors shall be less than 1 inch (25.4 mm) between resistor R1 and input pin, probes Y01-1 and Y01-2 and input and output pins, and load circuit and output pin

* FIGURE 13 Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for tables IX and X.

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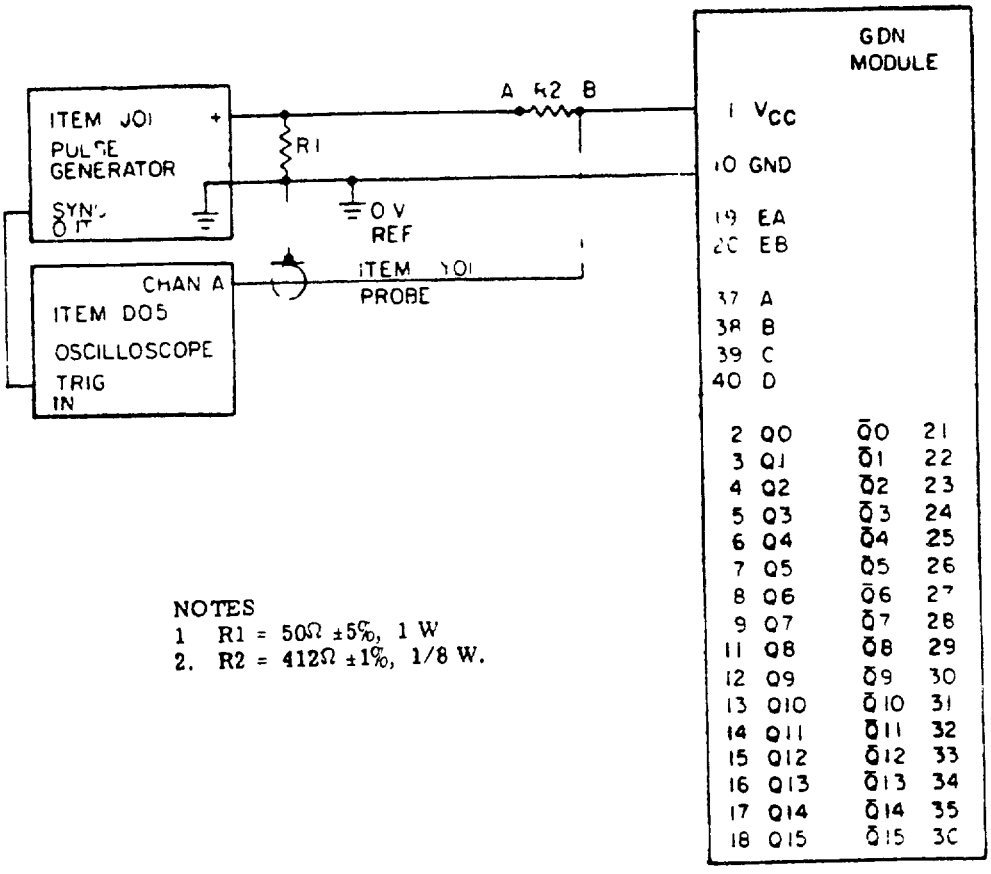
TABLE XI. Power supply transient rise time test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
P	Power supply transient rise time	2.5	7.5	µs	14,15

TABLE XII. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Q	Isolation, frame to module pins	All pins	Frame	10	∞	MΩ	16

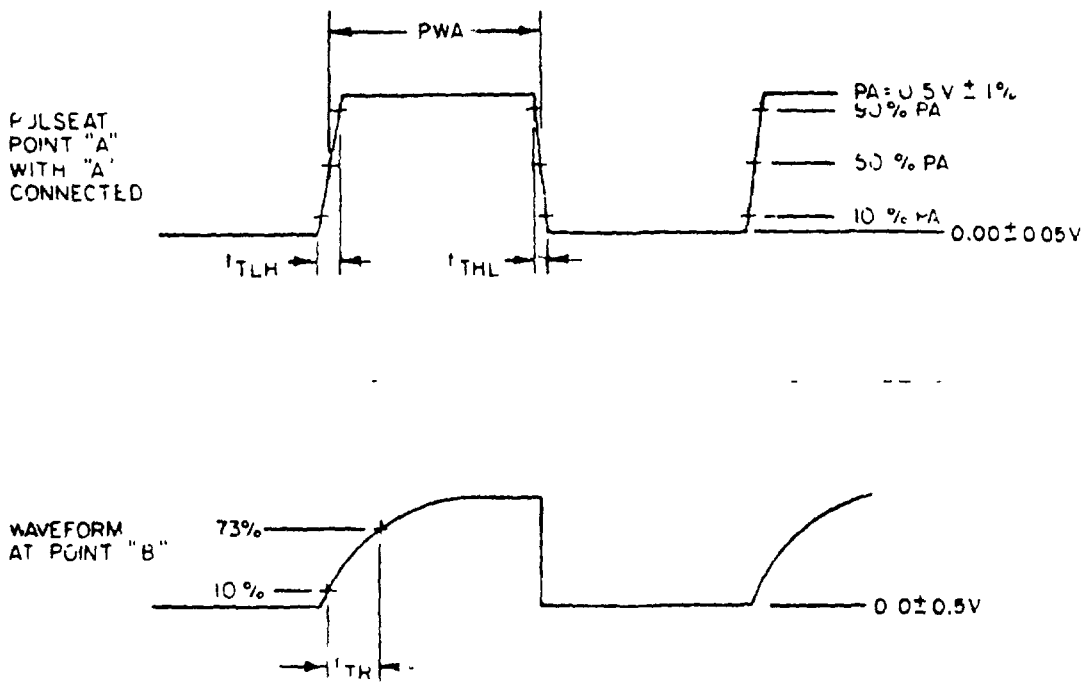
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NOTES
1. R1 = 50Ω ±5%, 1 W
2. R2 = 412Ω ±1%, 1/8 W.

* FIGURE 14. Power supply transient rise time test circuit for table VI.

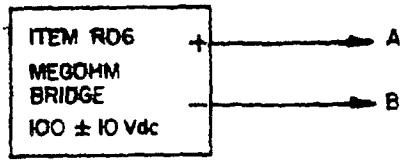
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NOTES
1. Input pulse parameters
   PWA = 30 µs ±10%.
   $t_{TLH} = t_{THL}$ = 50 ns ±10%.
   PRP = 60 µs ±10%.
2. Measure t_{TR} at point C.

* FIGURE 13. Power supply transient rise time test circuit waveforms for table XI.

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		GDN MODULE	
1	V_{CC}		
10	GND		
19	EA		
20	EB		
37	A		
38	B		
39	C		
40	D		
2	Q0	$\bar{Q}0$	21
3	Q1	$\bar{Q}1$	22
4	Q2	$\bar{Q}2$	23
5	Q3	$\bar{Q}3$	24
6	Q4	$\bar{Q}4$	25
7	Q5	$\bar{Q}5$	26
8	Q6	$\bar{Q}6$	27
9	Q7	$\bar{Q}7$	28
11	Q8	$\bar{Q}8$	29
12	Q9	$\bar{Q}9$	30
13	Q10	$\bar{Q}10$	31
14	Q11	$\bar{Q}11$	32
15	Q12	$\bar{Q}12$	33
16	Q13	$\bar{Q}13$	34
17	Q14	$\bar{Q}14$	35
18	Q15	$\bar{Q}15$	36

WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads

NOTE Penetrate the anodized finish or the module frame with a sharply pointed probe connected to the negative terminal

FIGURE 16. Isolation test circuit for table XII.

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* TABLE XIII. Output crosstalk (low-level and high-level) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	$\overline{Q}2$ Output crosstalk, low level	19,20,38,39, 40	37	23	Open	0	400	mV	17,18
R2	$\overline{Q}3$			24					
R3	$\overline{Q}4$			25					
R4	$\overline{Q}5$			26					
R5	$\overline{Q}6$			27					
R6	$\overline{Q}7$			28					
R7	$\overline{Q}8$			29					
R8	$\overline{Q}9$			30					
R9	$\overline{Q}10$			31					
R10	$\overline{Q}11$			32					
R11	$\overline{Q}12$			33					
R12	$\overline{Q}13$			34					
R13	$\overline{Q}14$			35					
R14	$\overline{Q}15$			36					
R15	$\overline{Q}0$	19,20,39,40		21	38				
R16	$\overline{Q}1$			22					
R17	$\overline{Q}4$			25					
R18	$\overline{Q}5$			26					
R19	$\overline{Q}6$			27					
R20	$\overline{Q}7$			28					
R21	$\overline{Q}8$			29					
R22	$\overline{Q}9$			30					
R23	$\overline{Q}10$			31					

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* TABLE XIII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R24	$\overline{Q}11$	Output crosstalk, low level	19,20,39,40	37	32	38	0	400	mV	17,18
R25	$\overline{Q}12$				33					
R26	$\overline{Q}13$				34					
R27	$\overline{Q}14$				35					
R28	$\overline{Q}15$				36					
R29	$\overline{Q}0$		19,20,38,40		21	39				
R30	$\overline{Q}1$				22					
R31	$\overline{Q}2$				23					
R32	$\overline{Q}3$				24					
R33	$\overline{Q}6$				27					
R34	$\overline{Q}7$				28					
R35	$\overline{Q}8$				29					
R36	$\overline{Q}9$				30					
R37	$\overline{Q}10$				31					
R38	$\overline{Q}11$				32					
R39	$\overline{Q}12$				33					
k40	$\overline{Q}13$				34					
R41	$\overline{Q}14$				35					
R42	$\overline{Q}15$				36					

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* TABLE XIII Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R43	$\overline{Q}0$	Output crosstalk, low level	19,20,40	37	21	38,39	0	400	mV	17,18
R44	$\overline{Q}1$				22					
R45	$\overline{Q}2$				23					
R46	$\overline{Q}3$				24					
R47	$\overline{Q}4$				25					
R48	$\overline{Q}5$				26					
R49	$\overline{Q}8$				29					
R50	$\overline{Q}9$				30					
R51	$\overline{Q}10$				31					
R52	$\overline{Q}11$				32					
R53	$\overline{Q}12$				33					
R54	$\overline{Q}13$				34					
R55	$\overline{Q}14$				35					
R56	$\overline{Q}15$				36					
R57	$\overline{Q}0$		19,20,38,39		21	40				
R58	$\overline{Q}1$				22					
R59	$\overline{Q}2$				23					
R60	$\overline{Q}3$				24					
R61	$\overline{Q}4$				25					
R62	$\overline{Q}5$				26					
R63	$\overline{Q}6$				27					
R64	$\overline{Q}7$				28					
R65	$\overline{Q}10$				31					
R66	$\overline{Q}11$				32					
R67	$\overline{Q}12$				33					

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* TABLE XIII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R68	$\overline{Q}13$	Output crosstalk, low level	19,20,38,39	37	34	40	0	400	mV	17,18
R69	$\overline{Q}14$		19,20,38,39		35	40				
R70	$\overline{Q}15$		19,20,38,39		36	40				
R71	$\overline{Q}0$		19,20,39		21	38,40				
R72	$\overline{Q}1$				22					
R73	$\overline{Q}2$				23					
R74	$\overline{Q}3$				24					
R75	$\overline{Q}4$				25					
R76	$\overline{Q}5$				26					
R77	$\overline{Q}6$				27					
R78	$\overline{Q}7$				28					
R79	$\overline{Q}8$				29					
R80	$\overline{Q}9$				30					
R81	$\overline{Q}12$				33					
R82	$\overline{Q}13$				34					
R83	$\overline{Q}14$				35					
R84	$\overline{Q}15$				36					
R85	$\overline{Q}0$		19,20,38		21	39,40				
R86	$\overline{Q}1$				22					
R87	$\overline{Q}2$				23					
R88	$\overline{Q}3$				24					
R89	$\overline{Q}4$				25					
R90	$\overline{Q}5$				26					

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* TABLE XIII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R91	$\overline{Q}$6 Output crosstalk, low level	19,20,38	37	27	39,40	0	400	mV	17,18
R92	$\overline{Q}$7			28					
R93	$\overline{Q}$8			29					
R94	$\overline{Q}$9			30					
R95	$\overline{Q}$10			31					
R96	$\overline{Q}$11			32					
R97	$\overline{Q}$14			35					
R98	$\overline{Q}$15			36					
R99	$\overline{Q}$0	19,20		21	38,39,40				
R100	$\overline{Q}$1			22					
R101	$\overline{Q}$2			23					
R102	$\overline{Q}$3			24					
R103	$\overline{Q}$4			25					
R104	$\overline{Q}$5			26					
R105	$\overline{Q}$6			27					
R106	$\overline{Q}$7			28					
R107	$\overline{Q}$8			29					
R108	$\overline{Q}$9			30					
R109	$\overline{Q}$10			31					
R110	$\overline{Q}$11			32					
R111	$\overline{Q}$12			33					
R112	$\overline{Q}$13			34					

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* TABLE XIII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	Q2	Output crosstalk, high level	19,20,38,39, 40	37	4	Open	0	400	mV	17,18
S2	Q3				5					
S3	Q4				6					
S4	Q5				7					
S5	Q6				8					
S6	Q7				9					
S7	Q8				11					
S8	Q9				12					
S9	Q10				13					
S10	Q11				14					
S11	Q12				15					
S12	Q13				16					
S13	Q14				17					
S14	Q15				18					
S15	Q0		19,20,39,40		2	38				
S16	Q1				3					
S17	Q4				6					
S18	Q5				7					
S19	Q6				8					
S20	Q7				9					
S21	Q8				11					
S22	Q9				12					
S23	Q10				13					

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* TABLE XIII. Output crosstalk (low-level and high-level) test - Continued.

Test condition		Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S24	Q11	Output crosstalk, high level	19,20,39,40	37	14	38	0	400	mV	17,18
S25	Q12				15					
S26	Q13				16					
S27	Q14				17					
S28	Q15				18					
S29	Q0		19,20,38,40		2	39				
S30	Q1				3					
S31	Q2				4					
S32	Q3				5					
S33	Q6				8					
S34	Q7				9					
S35	Q8				11					
S36	Q9				12					
S37	Q10				13					
S38	Q11				14					
S39	Q12				15					
S40	Q13				16					
S41	Q14				17					
S42	Q15				18					

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* TABLE XIII. Output crosstalk (low-level and high-level) test - Continued.

Test condition		Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S43	Q0	Output crosstalk, high level	19,20,40	37	2	38,39	0	400	mV	17,18
S44	Q1				3					
S45	Q2				4					
S46	Q3				5					
S47	Q4				6					
S48	Q5				7					
S49	Q8				11					
S50	Q9				12					
S51	Q10				13					
S52	Q11				14					
S53	Q12				15					
S54	Q13				16					
S55	Q14				17					
S56	Q15				18					
S57	Q0		19,20,38,39		2	40				
S58	Q1				3					
S59	Q2				4					
S60	Q3				5					
S61	Q4				6					
S62	Q5				7					
S63	Q6				8					
S64	Q7				9					
S65	Q10				13					
S66	Q11				14					
S67	Q12				15					

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* TABLE XIII. Output crosstalk (low-level and high-level) test - Continued.

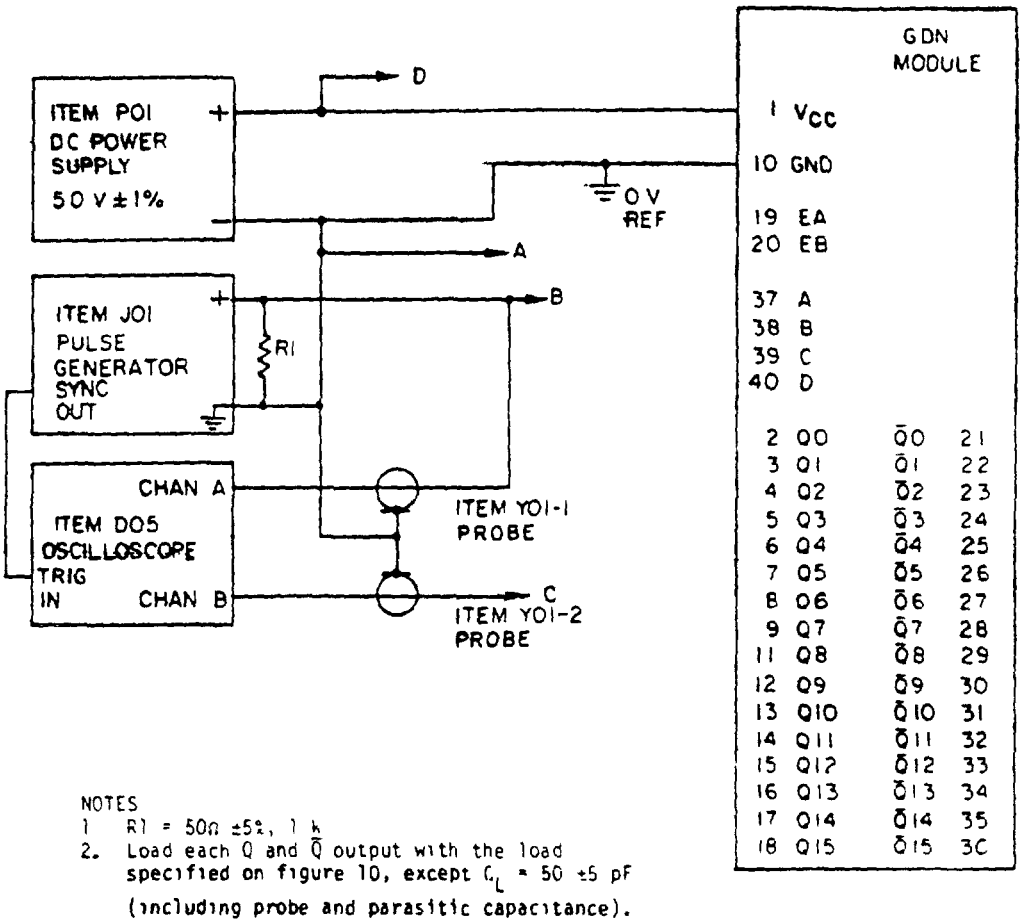
Test condition		Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S68	Q13	Output crosstalk, high level	19,20,38,39	37	16	40	0	400	mV	17,18
S69	Q14		19,20,38,39		17	40				
S70	Q15		19,20,38,39		18	40				
S71	Q0		19,20,39		2	38,40				
S72	Q1				3					
S73	Q2				4					
S74	Q3				5					
S75	Q4				6					
S76	Q5				7					
S77	Q6				8					
S78	Q7				9					
S79	Q8				11					
S80	Q9				12					
S81	Q12				15					
S82	Q13				16					
S83	Q14				17					
S84	Q15				18					
S85	Q0		19,20,38		2	39,40				
S86	Q1				3					
S87	Q2				4					
S88	Q3				5					
S89	Q4				6					
S90	Q5				7					

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* TABLE XIII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S91	Q6	Output crosstalk, high level	19,20,38	37	8	39,40	0	400	mV	17,18
S92	Q7				9					
S93	Q8				11					
S94	Q9				12					
S95	Q10				13					
S96	Q11				14					
S97	Q14				17					
S98	Q15				18					
S99	Q0		19,20		2	38,39,40				
S100	Q1				3					
S101	Q2				4					
S102	Q3				5					
S103	Q4				6					
S104	Q5				7					
S105	Q6				8					
S106	Q7				9					
S107	Q8				11					
S108	Q9				12					
S109	Q10				13					
S110	Q11				14					
S111	Q12				15					
S112	Q13				16					

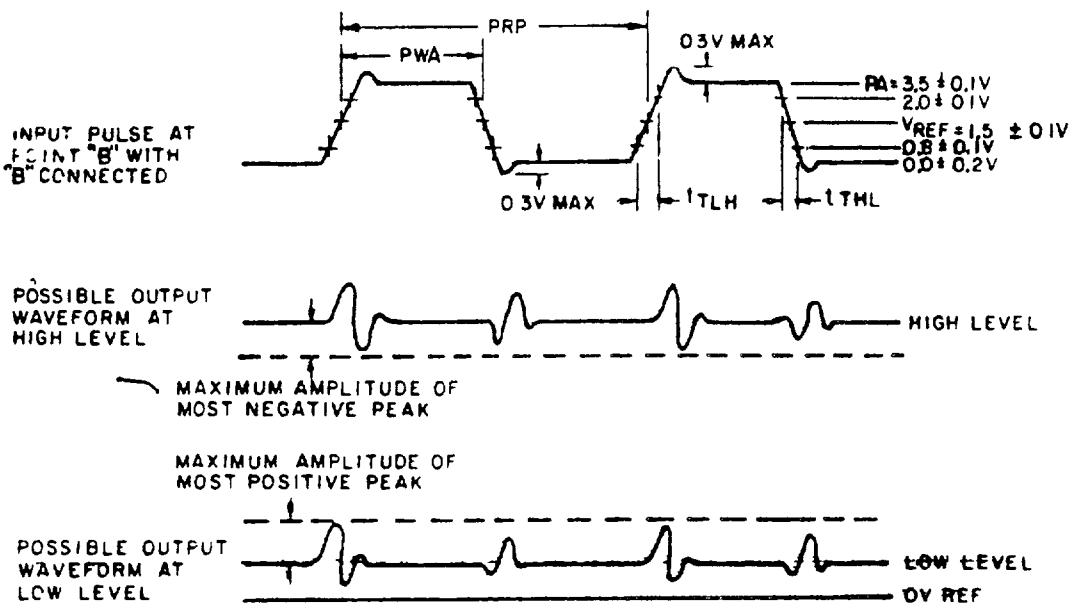
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NOTES
1. R1 = 50Ω ±5%, 1 k
2. Load each Q and $\bar{Q}$ output with the load specified on figure 10, except C_L = 50 ±5 pF (including probe and parasitic capacitance).

* FIGURE 17. Output crosstalk (low-level and high-level) test circuit for table XIII.

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NOTES

1 Input pulse parameters
tTHL = tTLH = 3.5 ±1.5 ns (verified at each input pin)
PWA = 200 ±20 ns
PRP = 1,000 ±100 ns

2 The lead length of electrical conductors shall be less than 2 inches (50.8 mm) between resistor R1 and input pins, probes Y01-1 and Y01-2 to input and output pins, load circuits to output pins, and ground leads of probes Y01-1 and Y01-2 to test fixture and module ground pins

* FIGURE 18. Output crosstalk (low-level and high-level) test circuit waveforms for table XIII.

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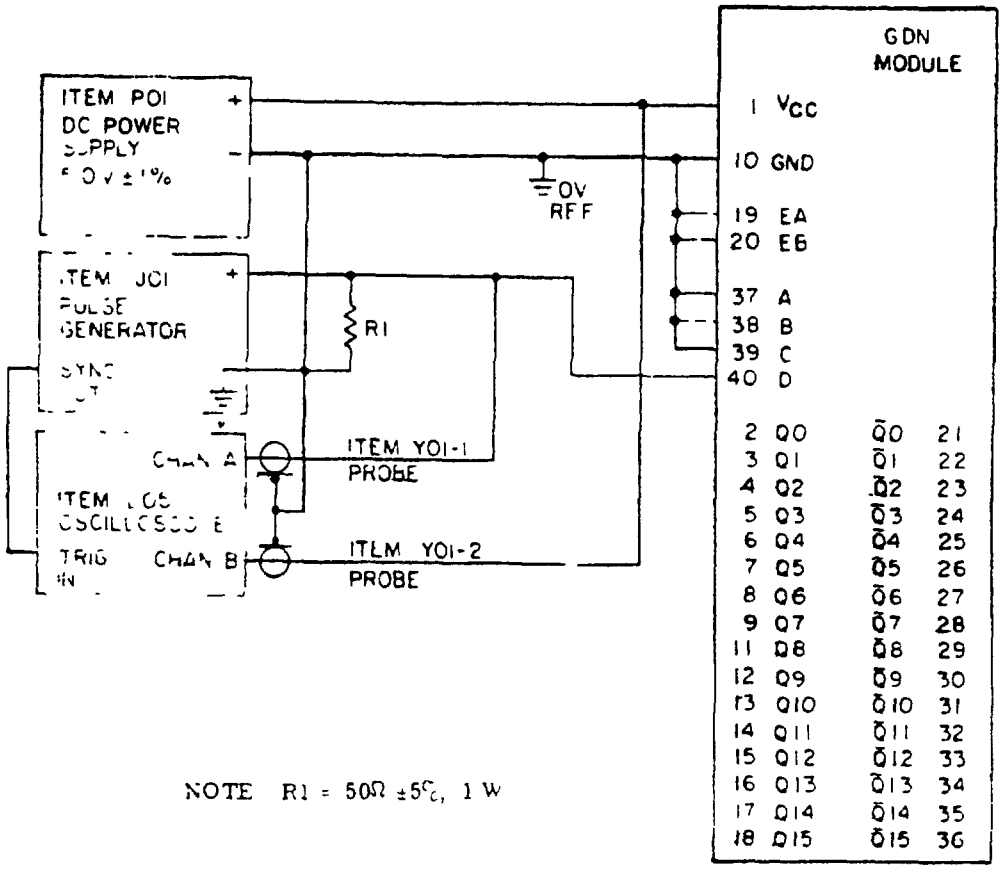
TABLE XIV. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
T	Power supply transient amplitude	0	400	mV	19,20

TABLE XV. Input capacitance test.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient		Unit	Test figure
				Min	Max		
U1	Enable A	Input capacitance	19	2	15	pF	21
U2	Enable B		20				
U3	A Input		37				
U4	B Input		38				
U5	C Input		39				
U6	D Input		40				

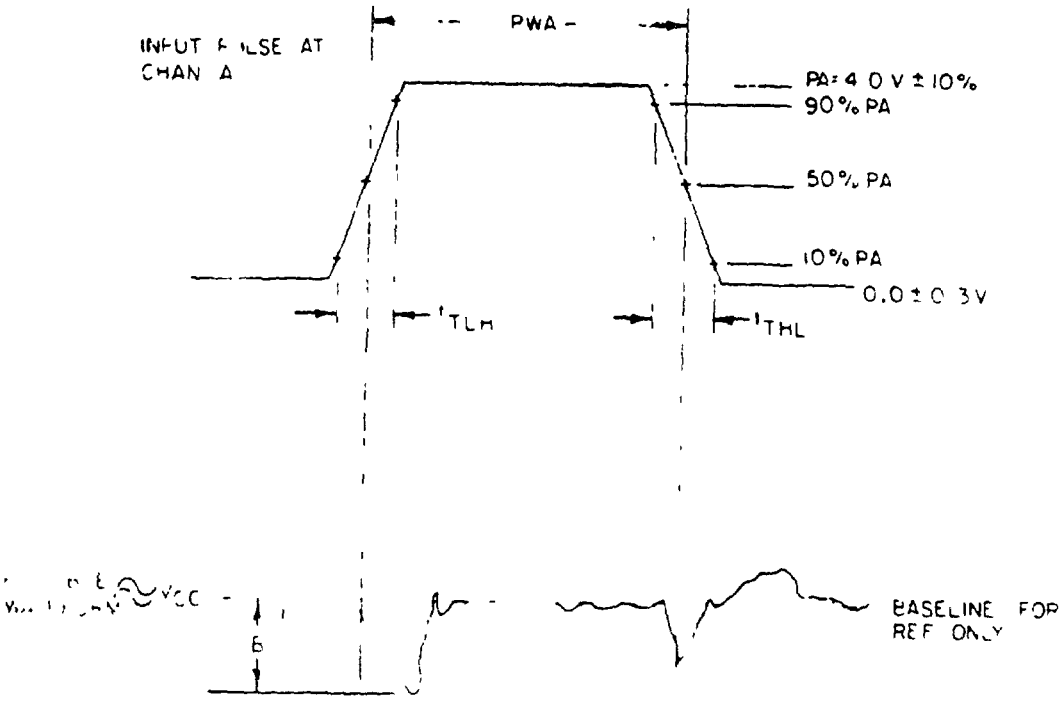
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NOTE R1 = 50Ω ±5%, 1 W

* FIGURE 19 Power supply transient amplitude test circuit for table XIV

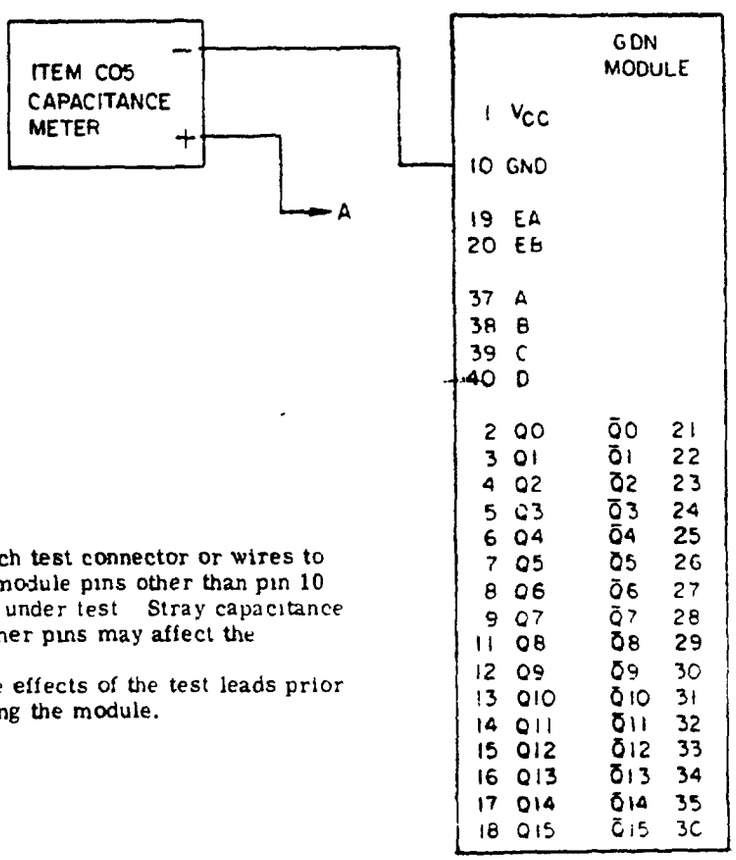
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NOTES
1. Input pulse parameters
   PWA = 3 µs ±10%.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   PRR = 100 kHz ±10%.
2. The maximum amplitude of the most negative spike (B) must comply with test condition T of table XIV.

* FIGURE 20 Power supply transient amplitude test circuit waveforms for table XIV

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NOTES

1. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
2. Null out the effects of the test leads prior to connecting the module.

FIGURE 21. Input capacitance test circuit for table I.

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INCH-POUND

MIL-M-28787/7C (SH)
AMENDMENT 4
18 December 1991
SUPERSEDING
AMENDMENT 3
11 December 1985

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

DECODER, BINARY, DIGITAL, KEY CODE GDN

This amendment forms a part of Military Specification MIL-M-28787/7C (EC), dated 7 September 1982, and is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense

PAGE 4

* TABLE I, Test conditions F1-F32, Test type column: Delete "X" and substitute "S", (two places)

PAGES 4, 5, AND 6

TABLE I, Table heading, Class II Delete "100 ± 5°C" (three places) and substitute "85 ± 5°C" (three places)

TABLE I, Test conditions F17-F32, Class II Min columns: Delete "30" (two places) and substitute "20" (two places)

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PAGE 5

TABLE I, Class II, 85 ± 5°C Max column

Test condition M1–M31 (odd numbers), Initial: Delete "52" and substitute "43"
End-of-life: Delete "57" and substitute "48".

Test condition M2–M32 (even numbers), Initial: Delete "75" and substitute "63"
End-of-life: Delete "80" and substitute "68"

Test condition N1–N31 (odd numbers), Initial Delete "47" and substitute "40"
End-of-life: Delete "52" and substitute "45".

Test condition N2–N32 (even numbers), Initial: Delete "70" and substitute "68"
End-of-life: Delete "75" and substitute "73"

PAGE 7

* 4 4.2 Delete and substitute the following.

"4 4.2 Order of testing Test conditions A through U shall be performed in any order with the following exceptions Test conditions A, B, C, F, and Q are stress tests and shall be performed only at 25 ± 5°C If test conditions A, B, C, F, or Q are part of a sequence of tests, the order of testing shall be as follows Q, A, B, C, and F where applicable, followed by the remaining tests in the sequence "

* 4 5 1, fifth and sixth lines Delete "C through Q" and substitute "D, E, G through P, and R through T".

* 4.5 3, third and fourth lines Delete "C through Q" and substitute "D, E, G through P, and R through T".

* 4.5 3, sixth line Delete "C through U" and substitute "D, E, G through P, and R through T"

PAGE 18

FIGURE 7. Delete R1 from the line from the negative terminal of ITEM P01 to A and place it in the line from the positive terminal of ITEM P01 to O V REF

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PAGE 24

* TABLE VII, Test condituon G33, Parameter column. Delete "$\overline{Q}$0" and substitute "Q0"

PAGE 36

* TABLE VII, Test condituon H9, F to pin column· Add pin "38".

PAGE 50

FIGURE 10, NOTE 1(e): Delete and substitute the following

"(e) R2 = Select value for a current flow of 14 4 mA ± 1% with point L held at 0.4 V."

FIGURE 10, NOTE 2(e): Delete and substitute the following.

"(e) R2 = Select value for a current flow of 16 mA ± 1% with point L held at 0.4 V "

FIGURE 15· Delete and substitute new figure 15 as printed on page 4 of this amendment

NOTE The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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